

Fundamental limits for thermodynamic control with quantum feedback

Kaiyuan Ji (Cornell), Gilad Gour (Technion), Mark M. Wilde (Cornell)



[arXiv:2503.09012](https://arxiv.org/abs/2503.09012)

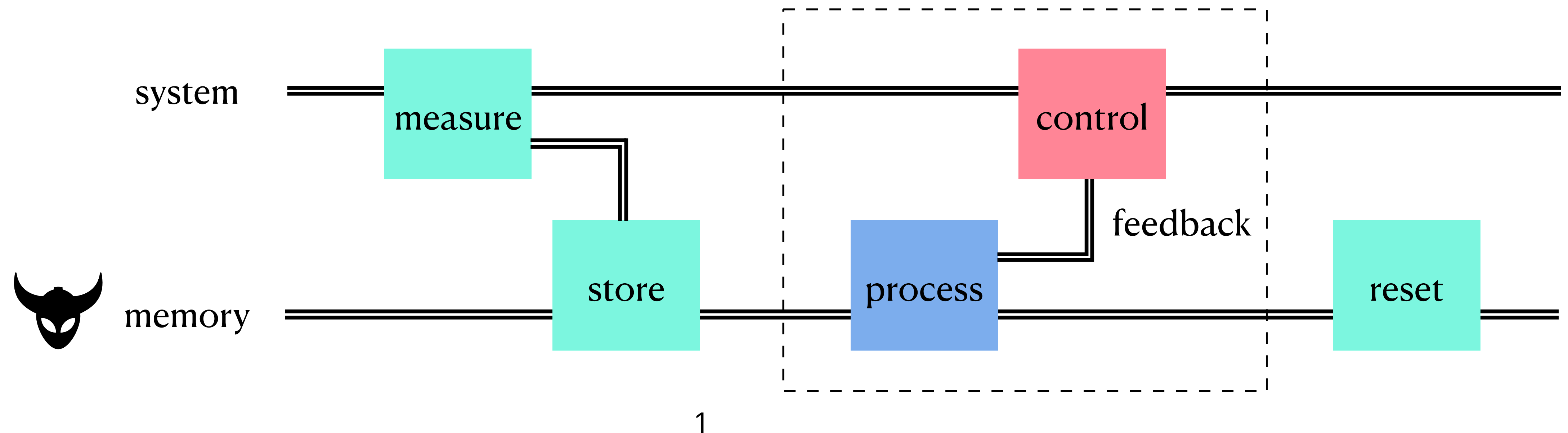
Beyond IID 2026, Shenzhen

Maxwell's demon & feedback control

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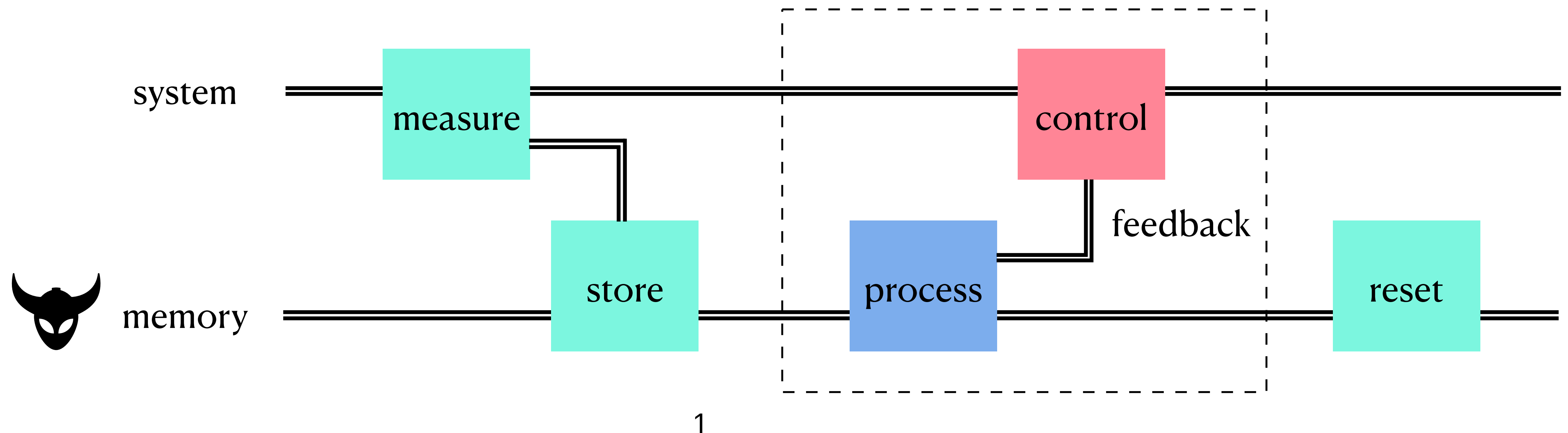
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$$\langle W \rangle \geq \Delta F_{\text{sys}} + \beta^{-1} \Delta I_{\text{classical}}$$

Parrondo et al., Nat. Phys. 11, 131 (2015)



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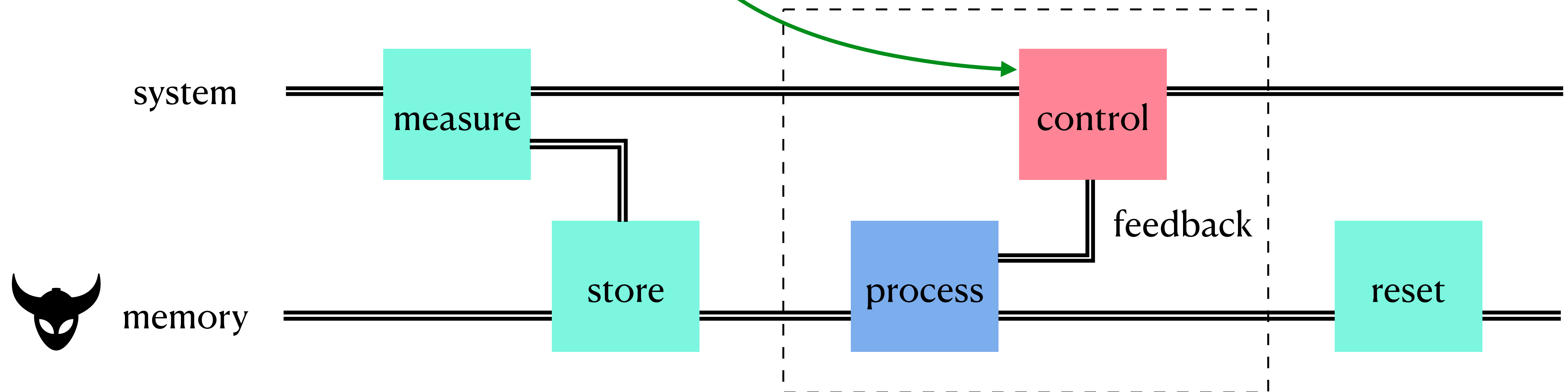
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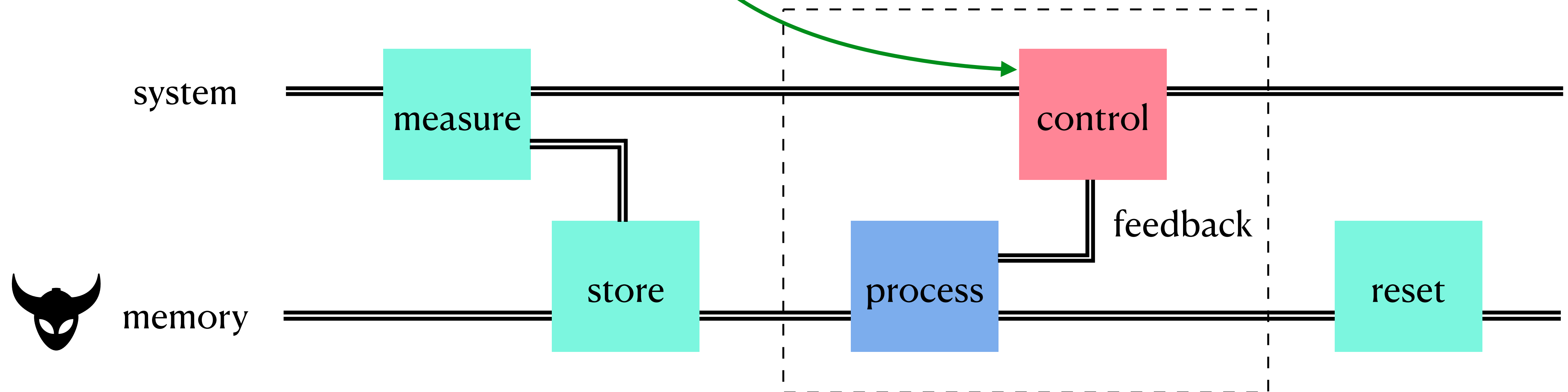
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feedback advantage
would be paid back in
measurement & reset

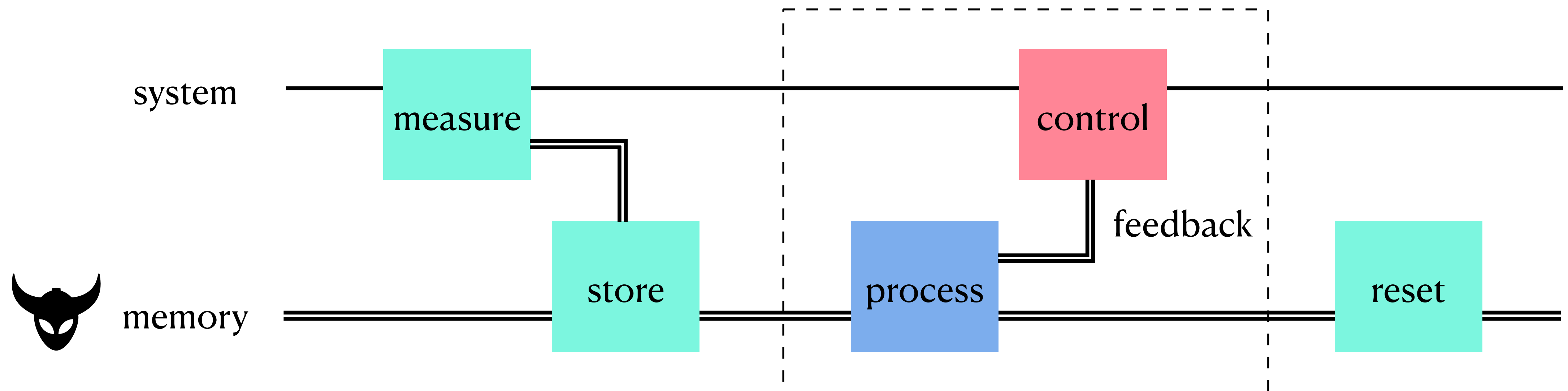
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Classical vs. quantum feedback

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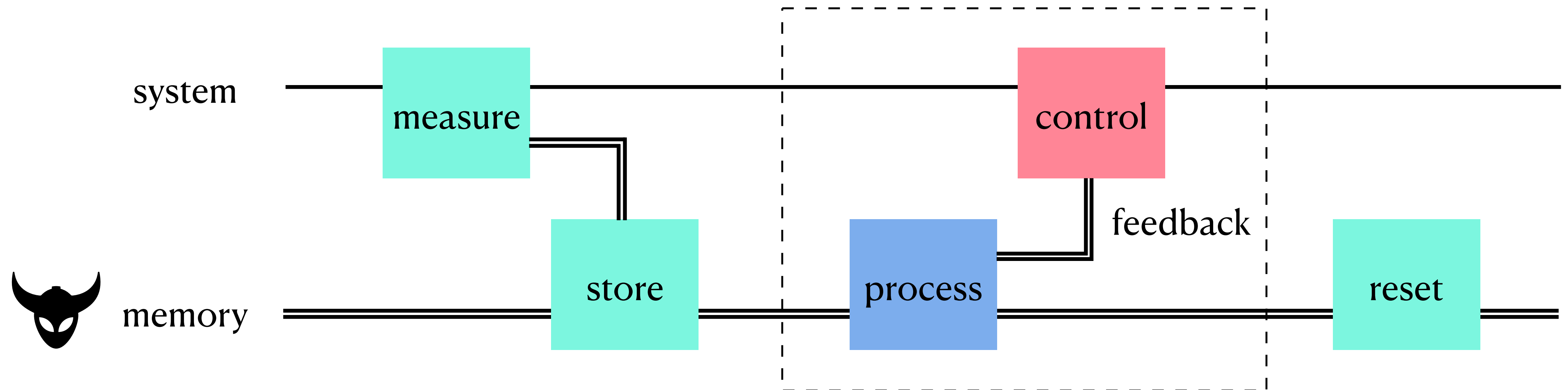


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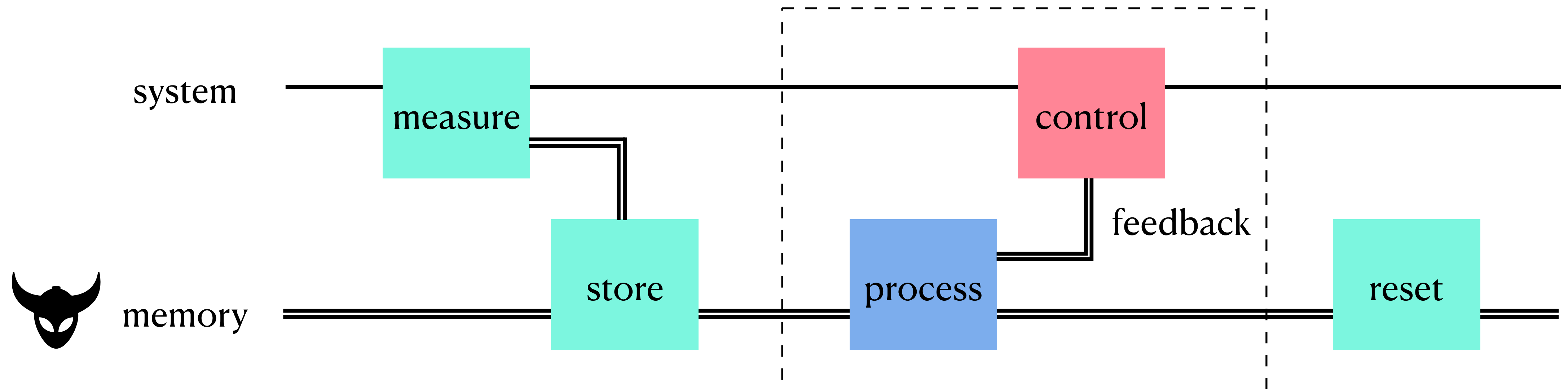


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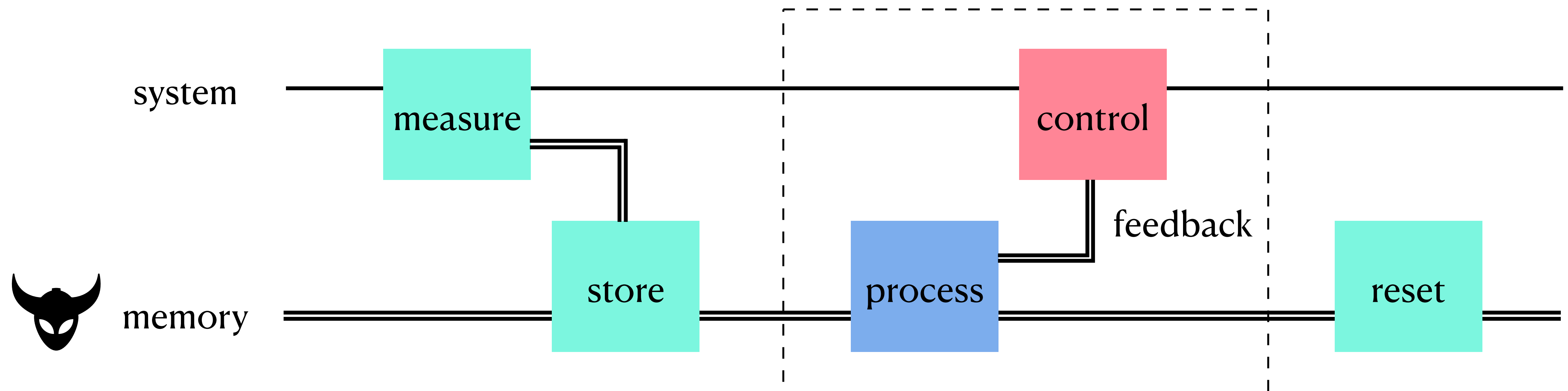


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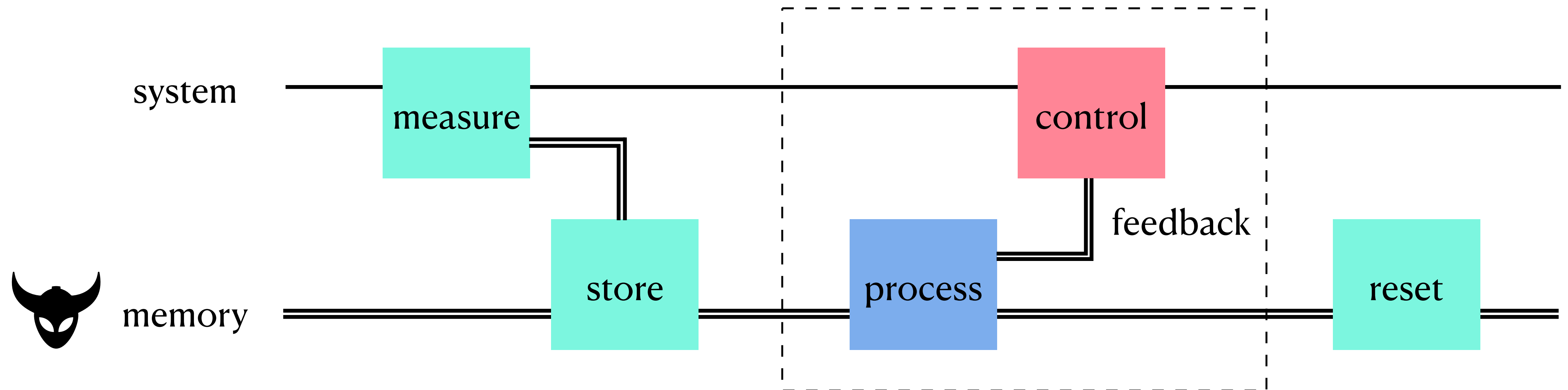
Generalizations to quantum systems & control were also studied

- But side information is always assumed to be collected by measurements
- Even weak or indirect measurements on quantum systems can only yield **classical** side information, which is then fed back classically
- Such assumptions do not capture the most general situation allowed by physics



Fully quantum feedback

A fully quantum feedback-controller would be able to acquire, process, and feed back **quantum** side information coherently

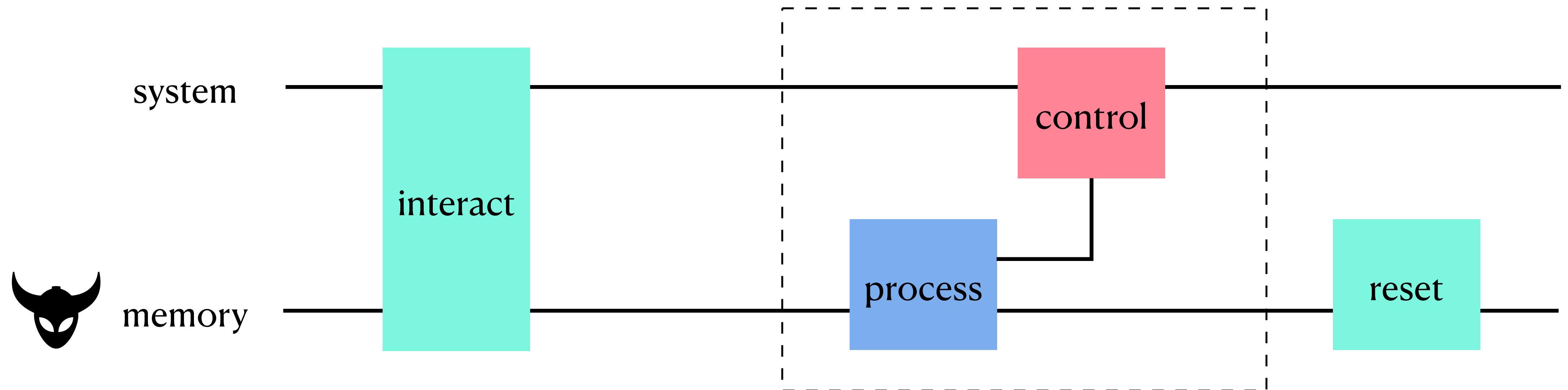


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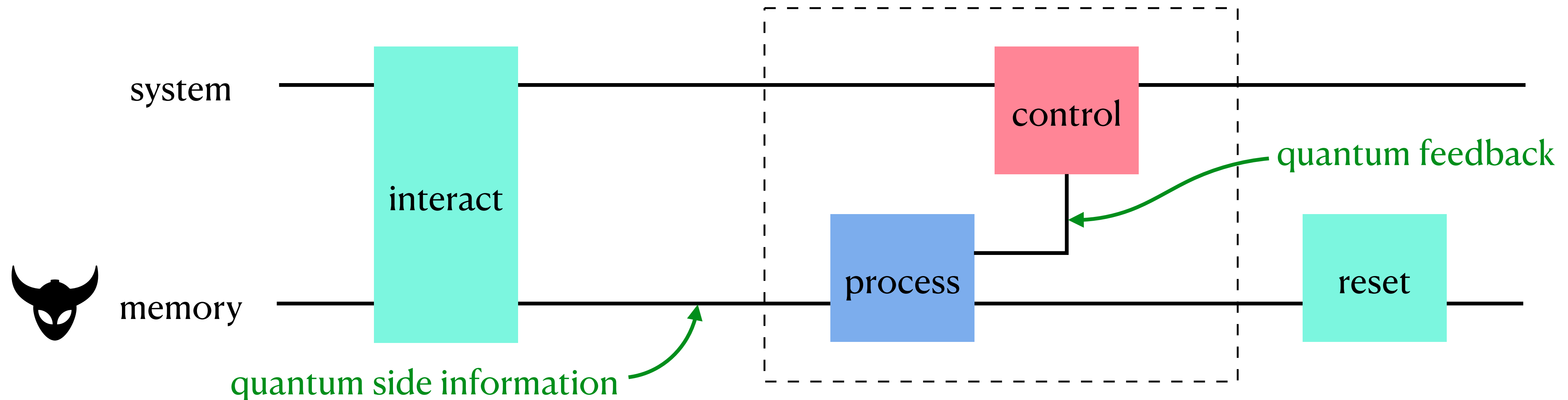
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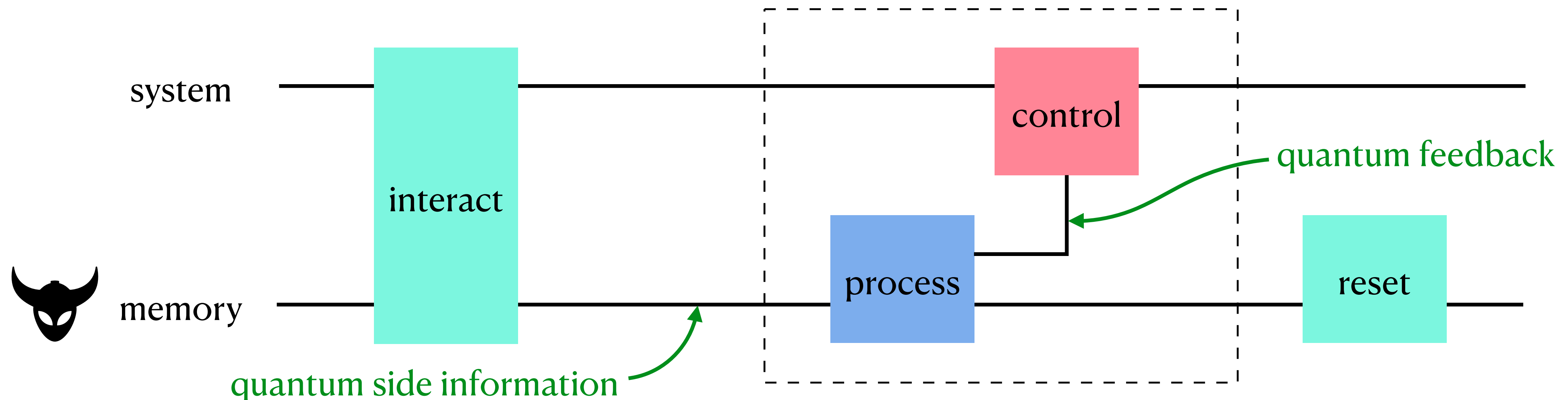


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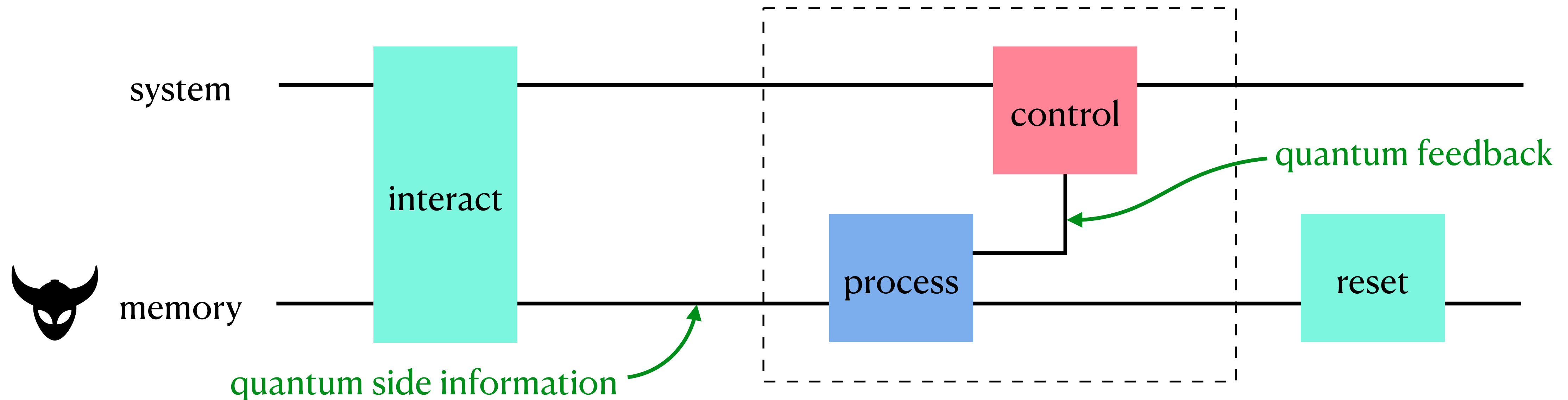
Seth Lloyd, PRA 62, 022108 (2000)

- Some tasks in quantum control theory, such as state exchange and entanglement transfer, are only achievable with quantum feedback; impossible with classical feedback
- Only bounds assuming quantum feedback qualify as “fundamental no-go limits” for all physically admissible feedback-control processes



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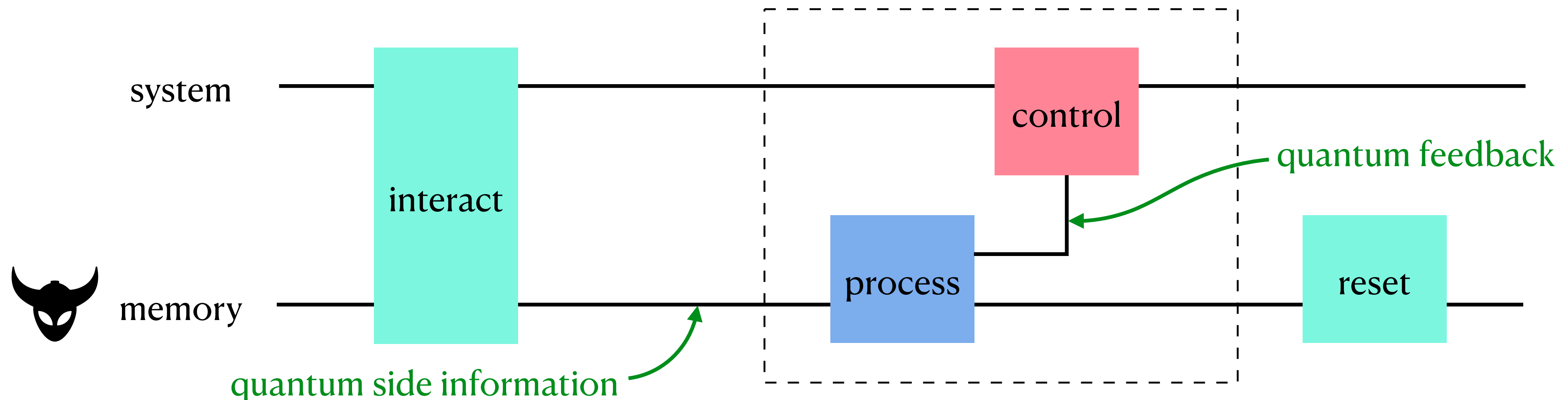


Fully quantum feedback

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del Rio et al., Nature 474, 61 (2011)

- May lead to exotic thermodynamic effects (such as cooling during erasure) due to negativity of quantum conditional entropy



Question:

How much work is fundamentally required for, or extractable from, a conversion when a controller feeds back quantum side information coherently into a thermodynamic system?

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- under fully quantum feedback control
- on both macroscopic (asymptotic) and microscopic (one-shot) scales

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Need a precise description for microscopic thermodynamic operations with quantum feedback

- Motivates us to define a resource theory of thermodynamics with quantum side information

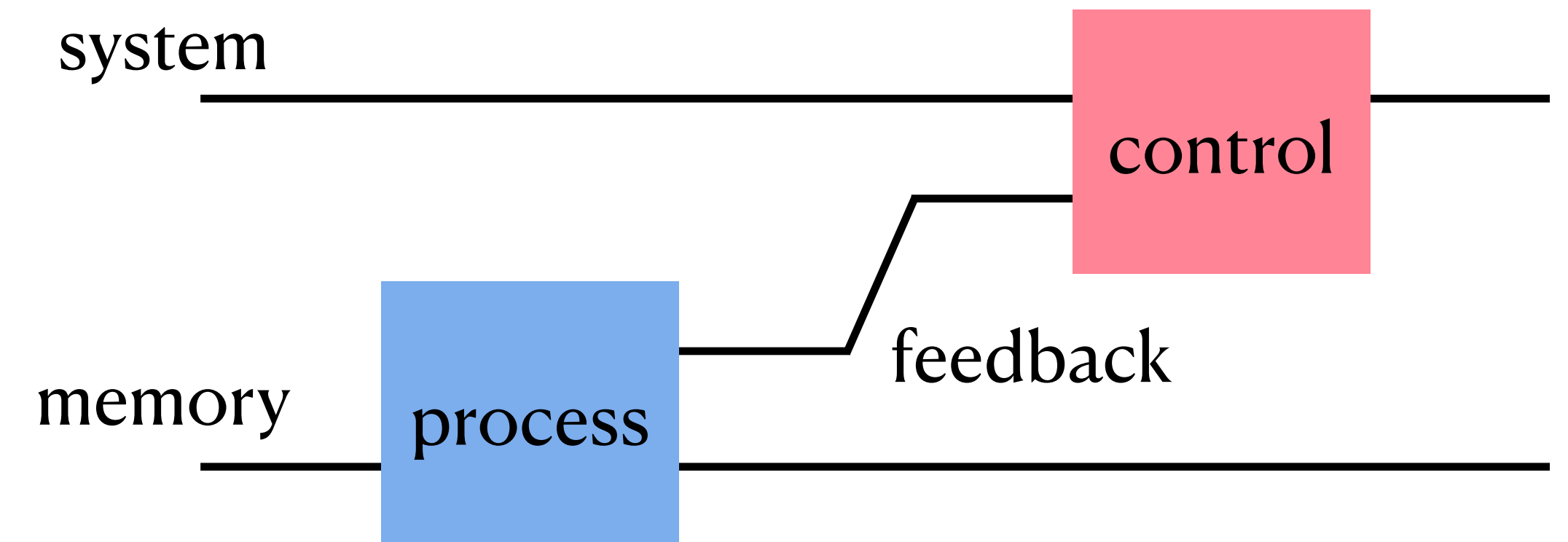
Operational framework:

Q. thermodynamics in the presence of q. side information

Free operations

We only focus on feedback control per se

Everything before that (e.g., acquisition of q. side info.) is encapsulated in the input state

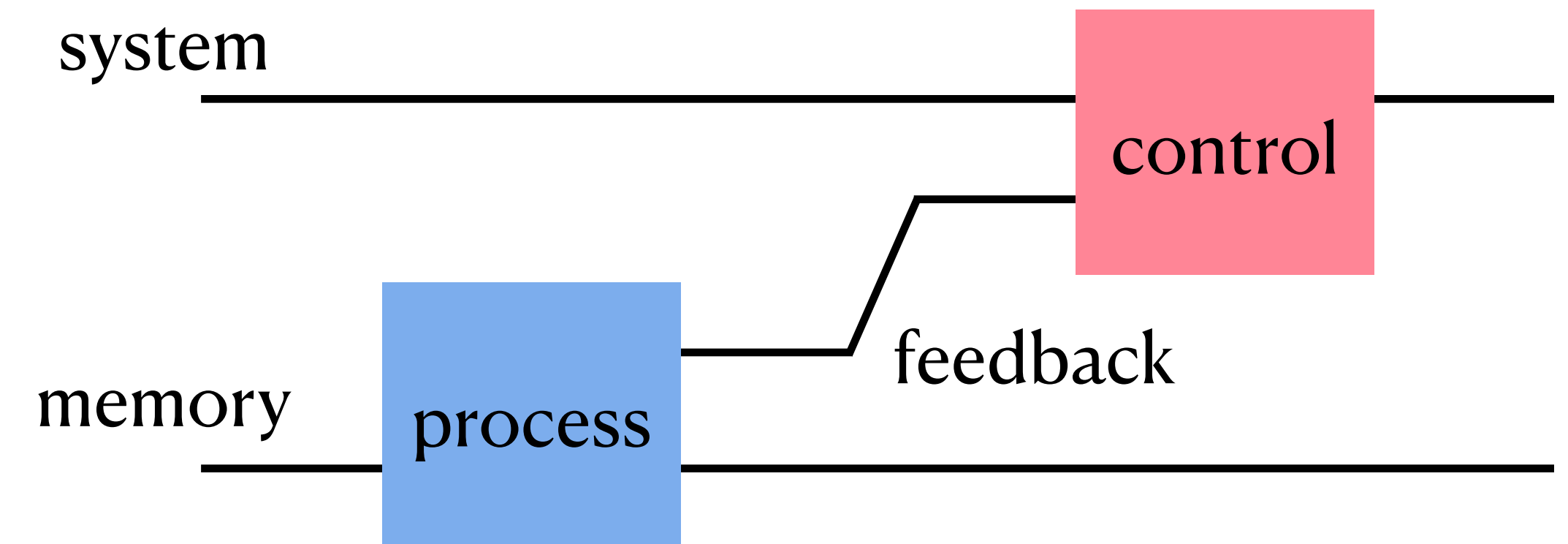


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Axiomatic approach: every physically admissible feedback-control scheme that can be implemented without supplying work to the system should be free



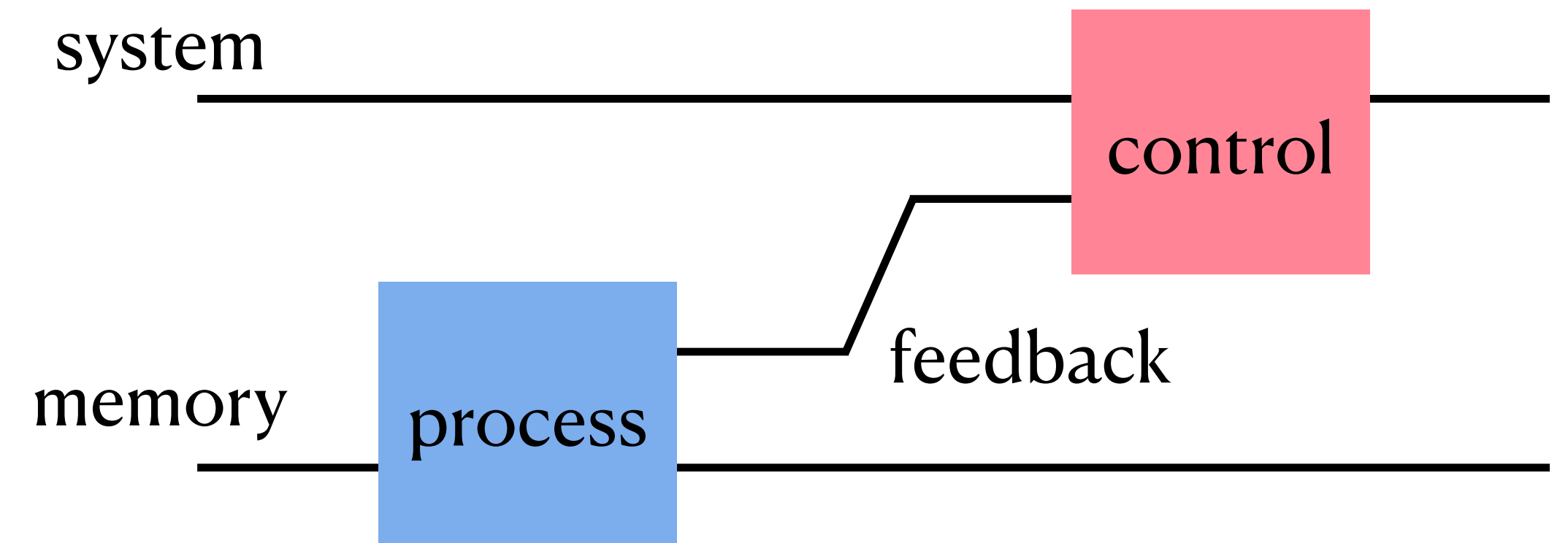
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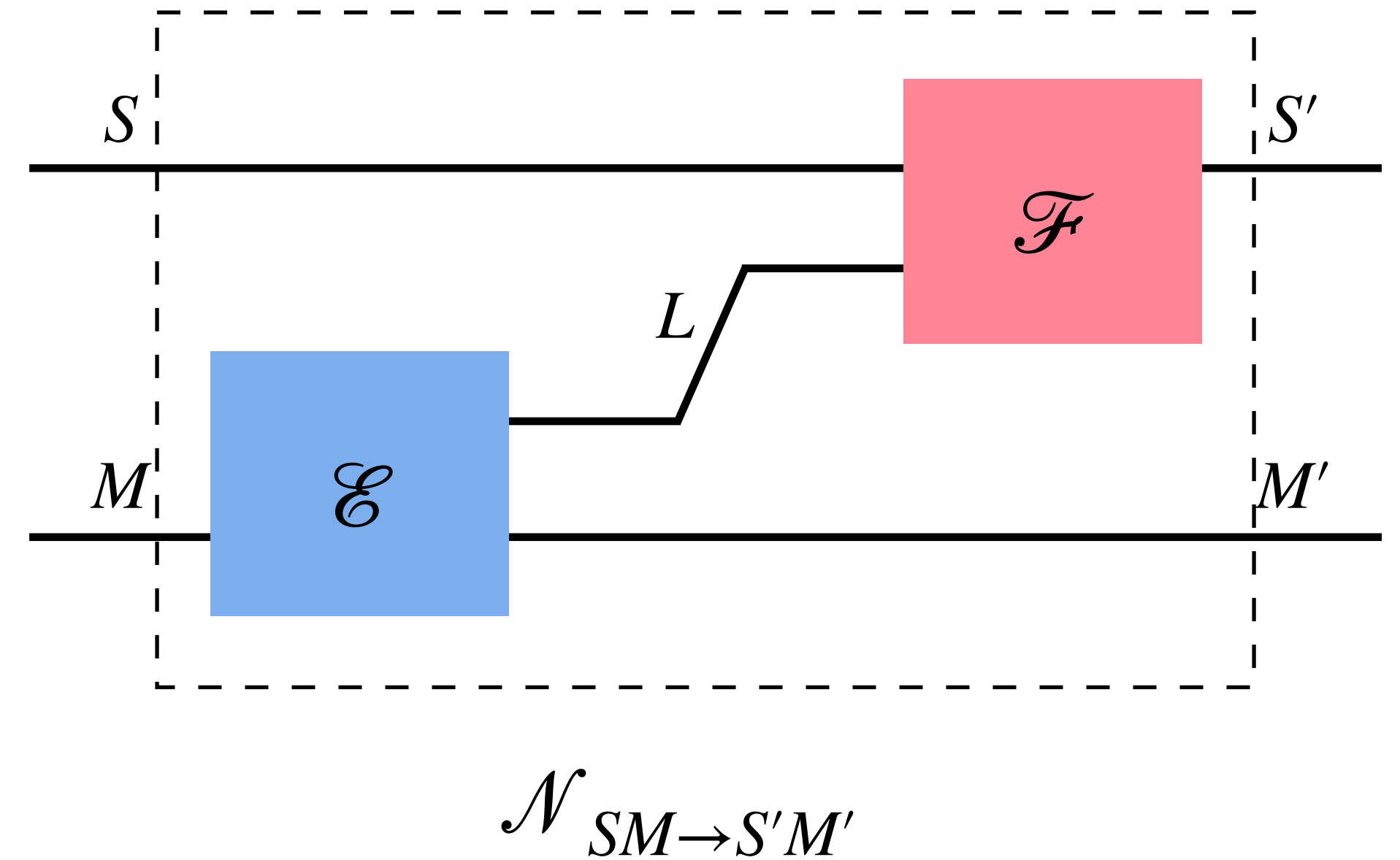
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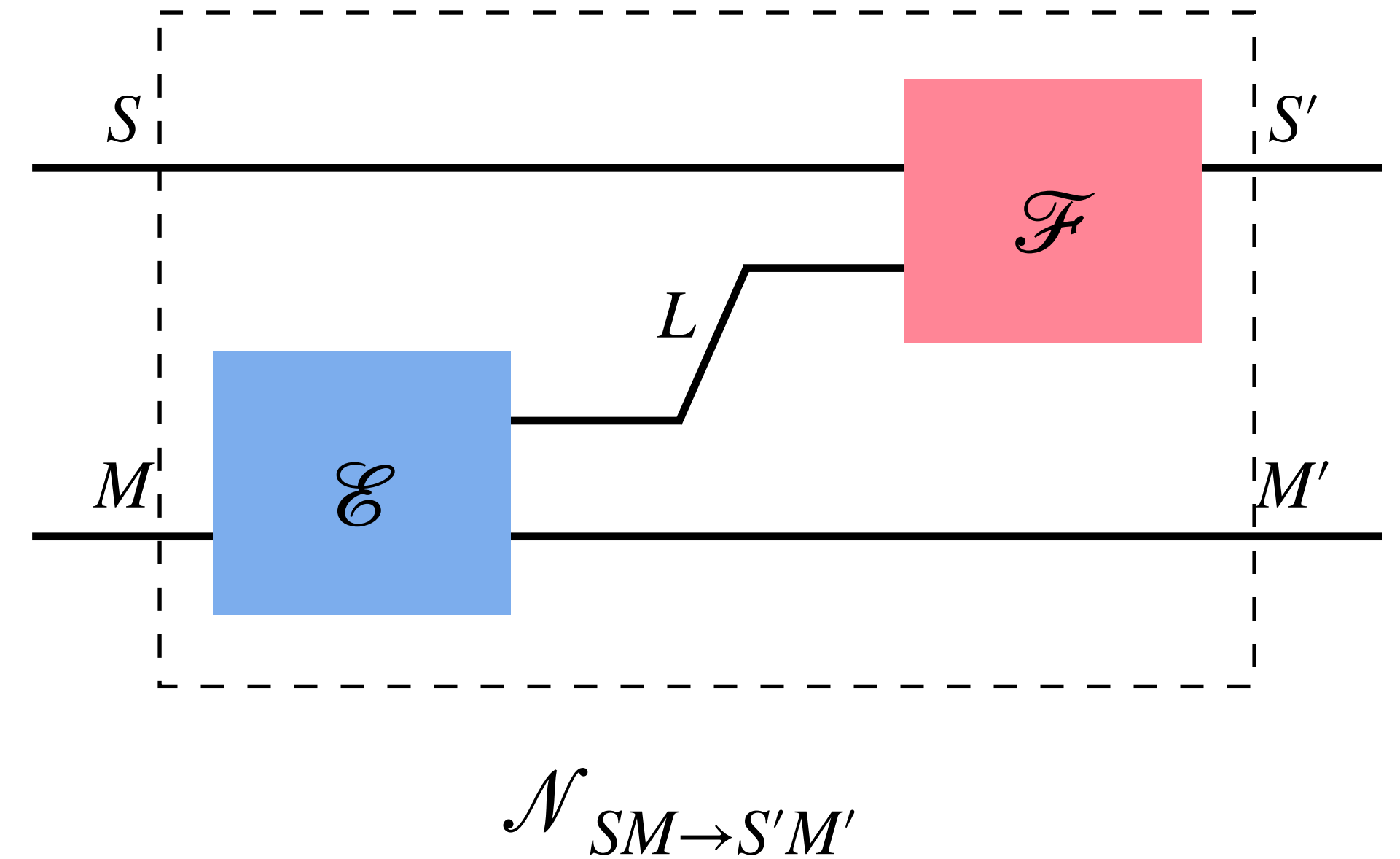
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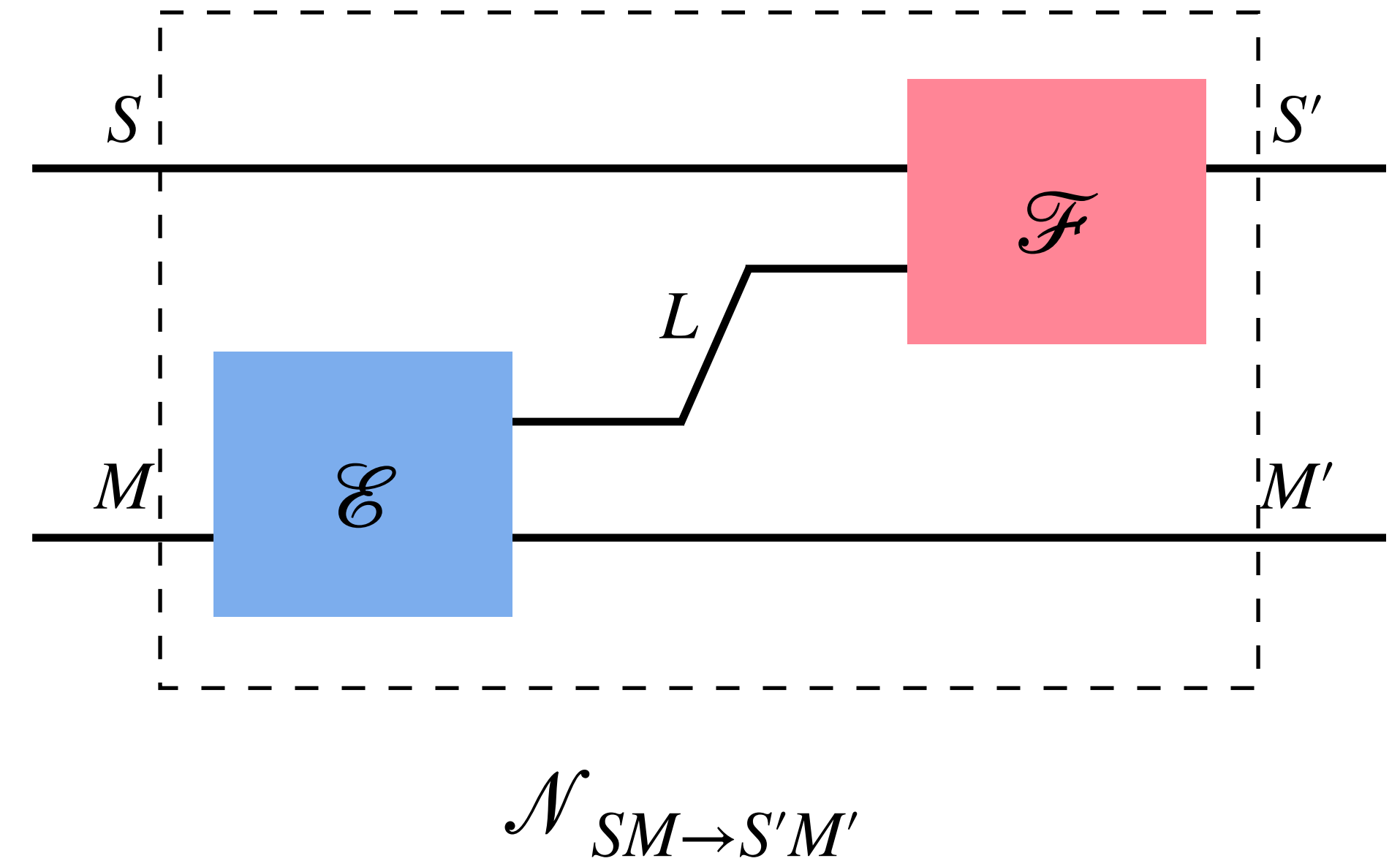
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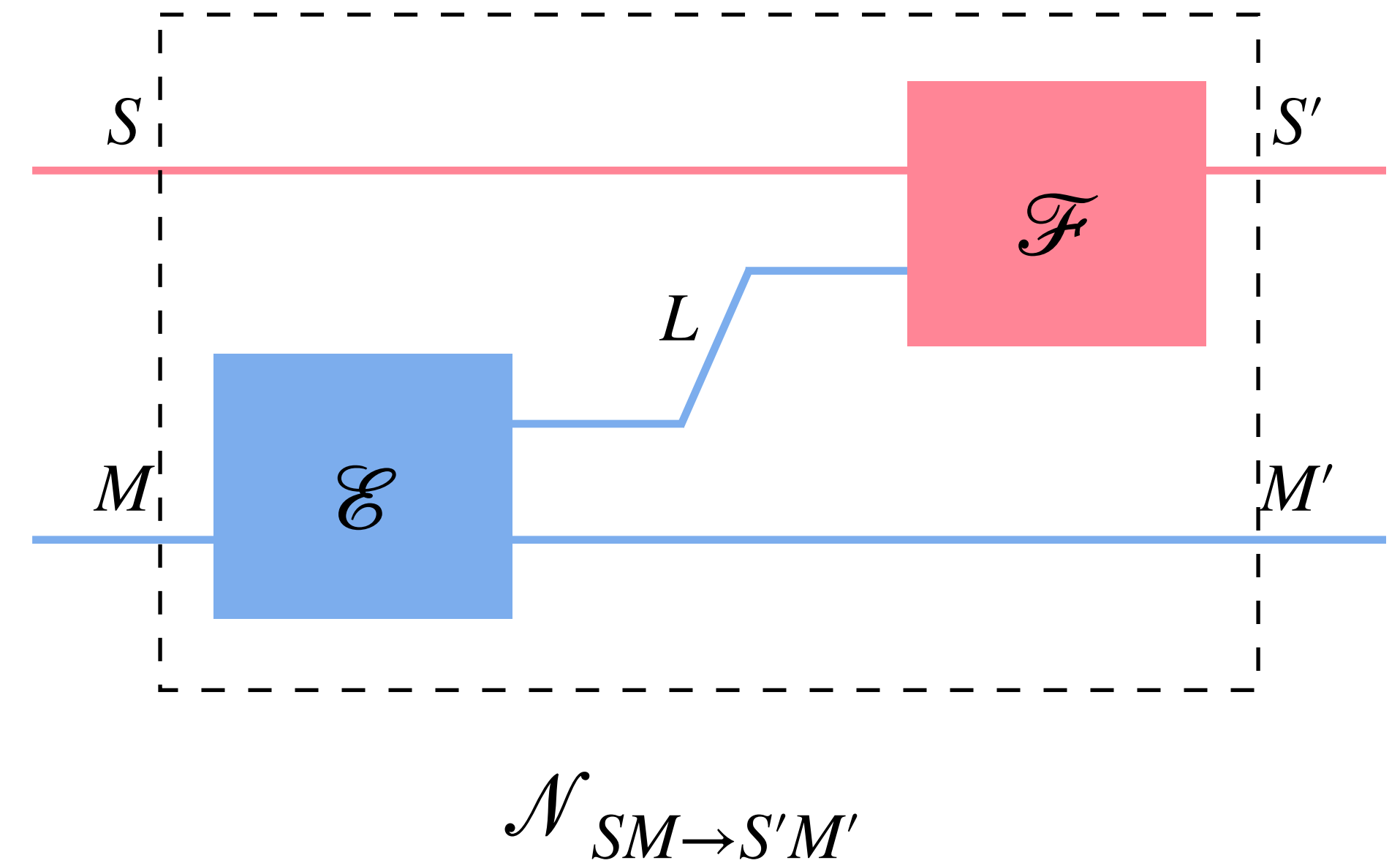
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$$\mathcal{F}_{SL \rightarrow S'}[\gamma_S \otimes \rho_L] = \gamma_{S'} \quad \forall \rho_L \quad \text{--- Gibbs states}$$



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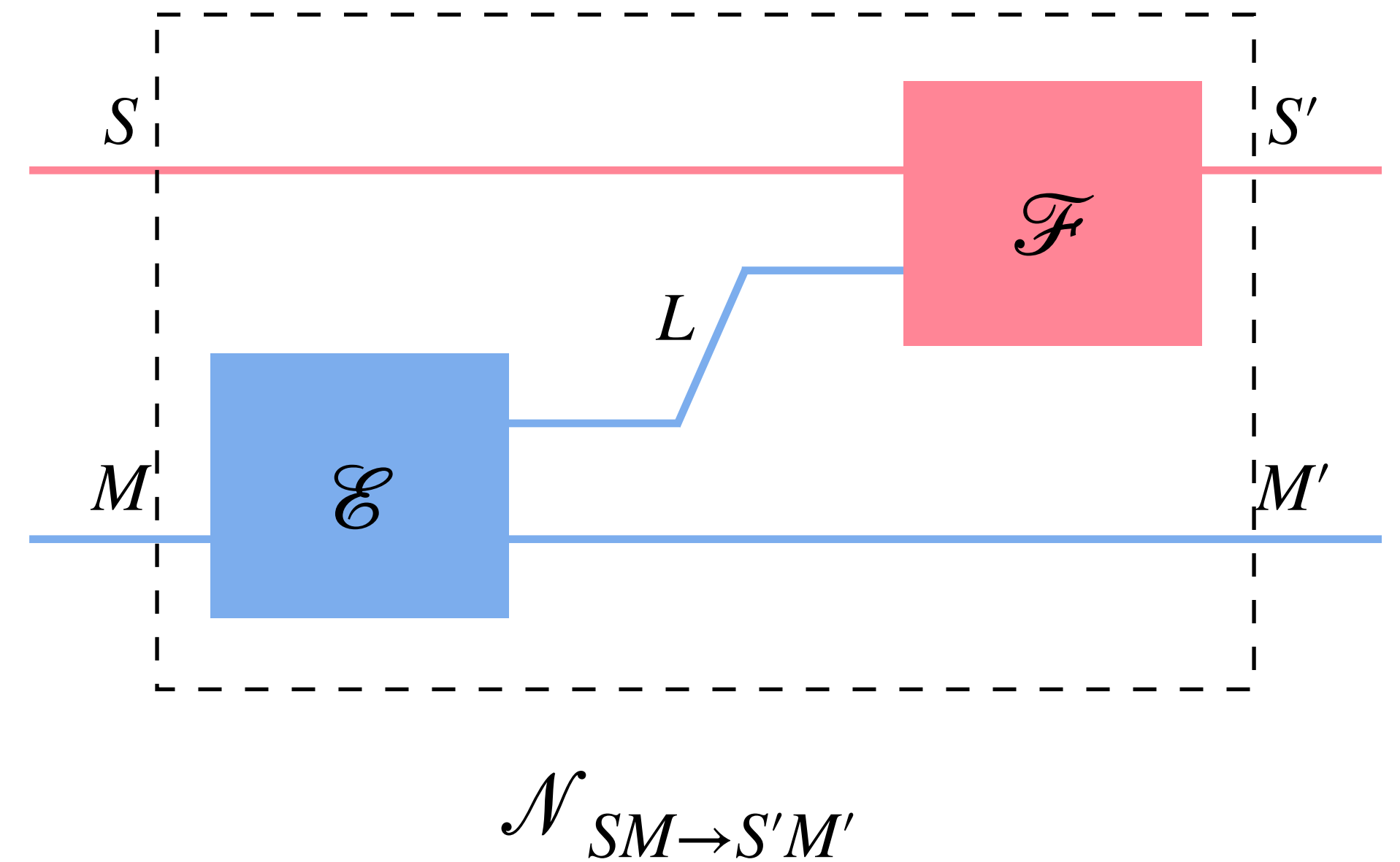
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feedback (L) only supplies information, not work (athermality resource)



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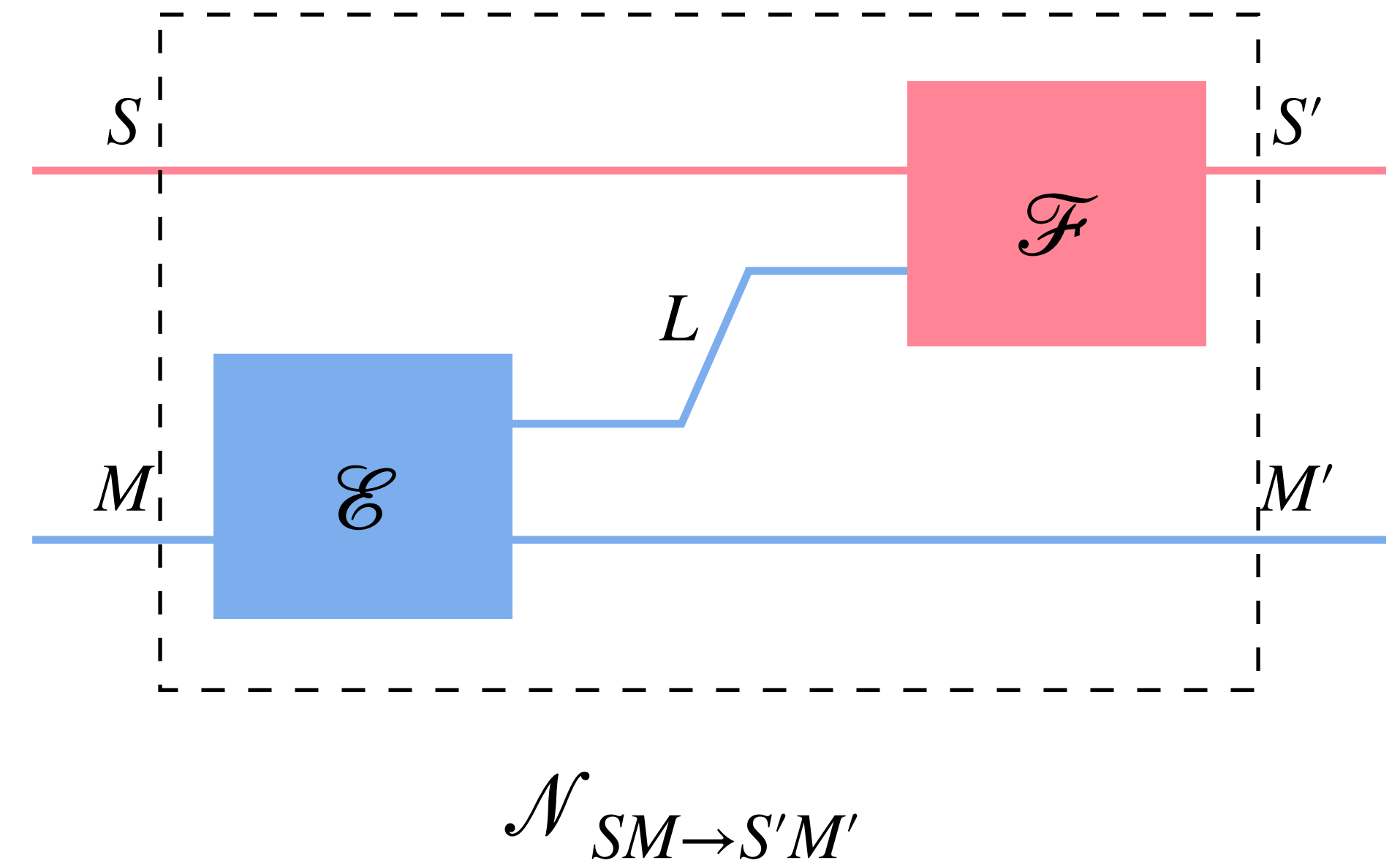
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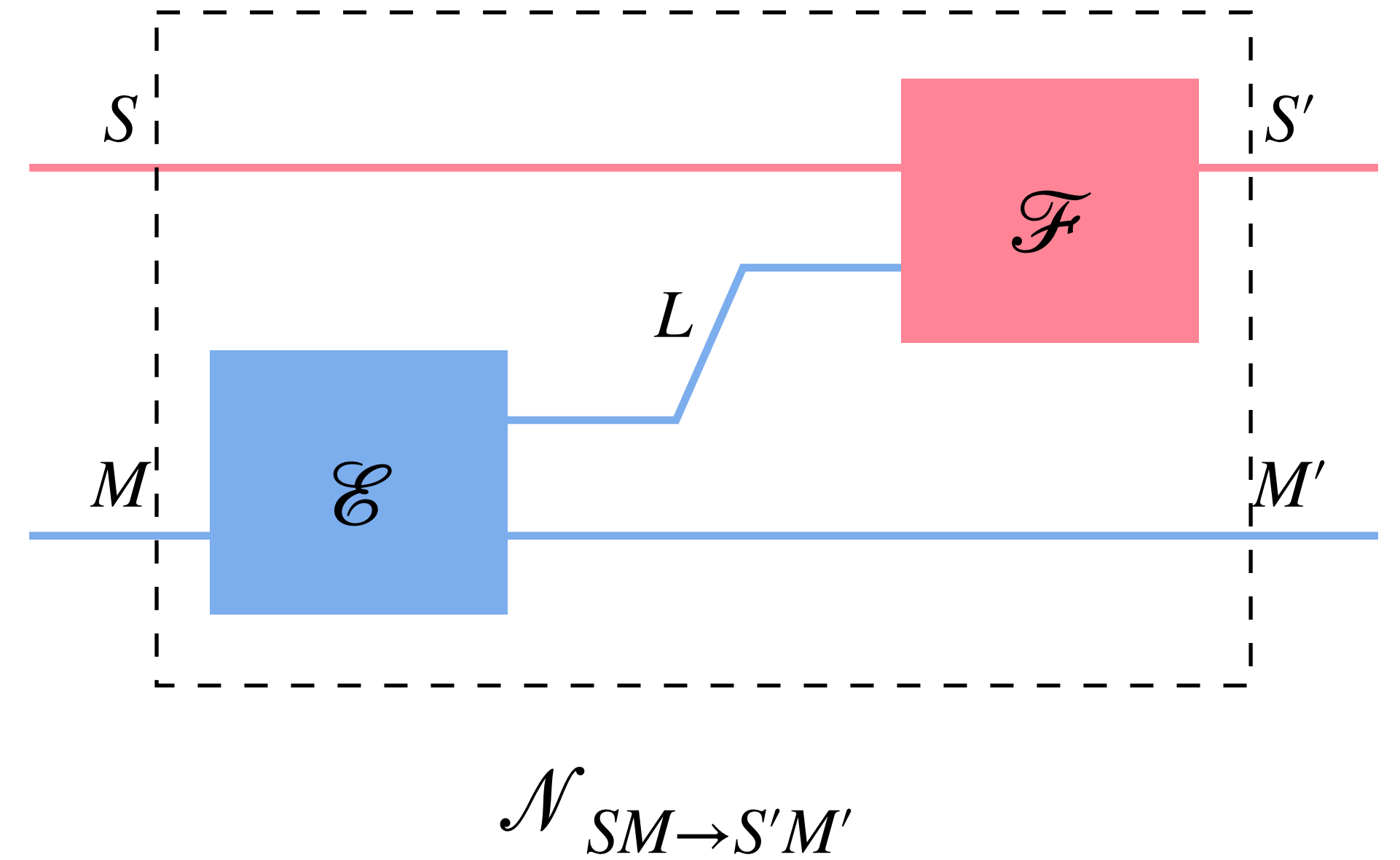
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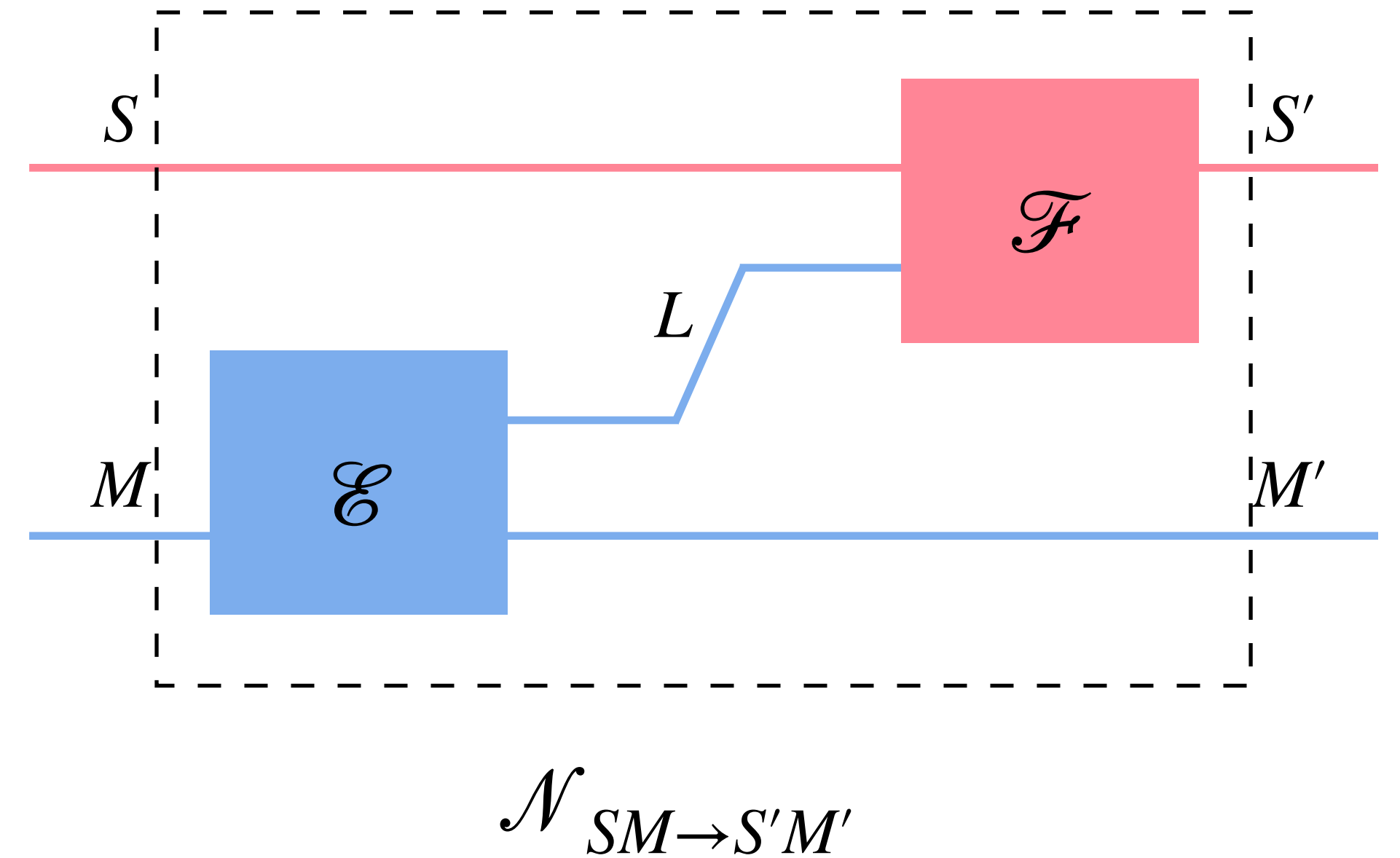
“Quantum-feedback-assisted Gibbs-preserving operations” (QFGOs)



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Free operations

“Quantum-feedback-assisted Gibbs-preserving operations” (QFGOs)



QFGOs are deliberately permissive, so our no-go limits apply universally

If something is impossible under QFGOs, then it is impossible via any feedback-control scheme regardless of operational assumptions

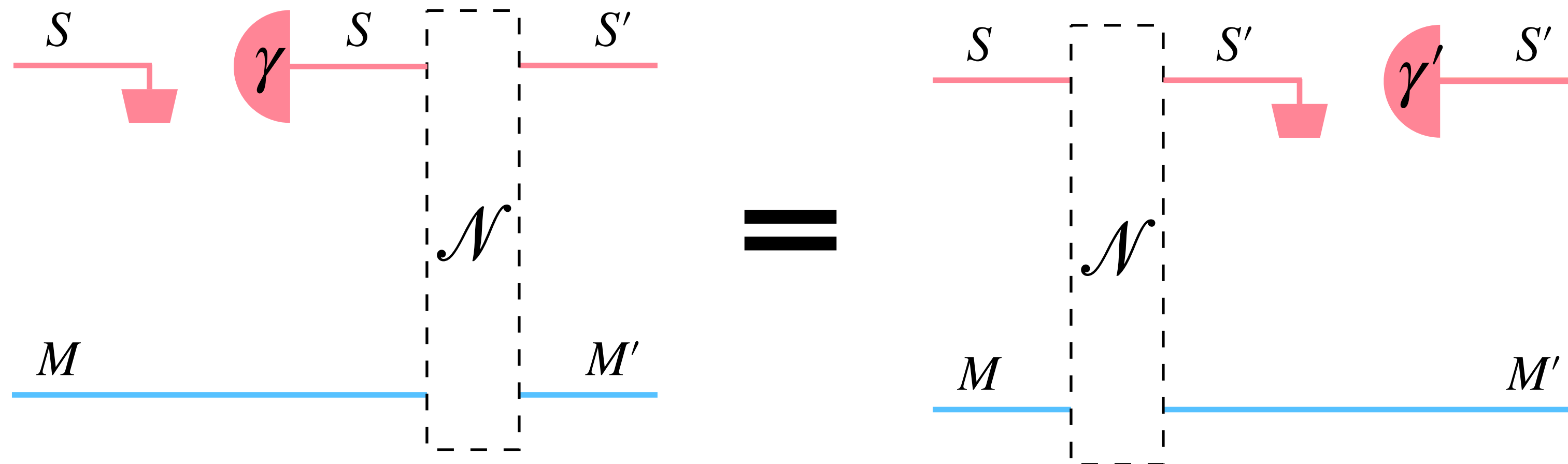
QFGOs: equivalent characterization

Lemma. (Thermalization covariance)

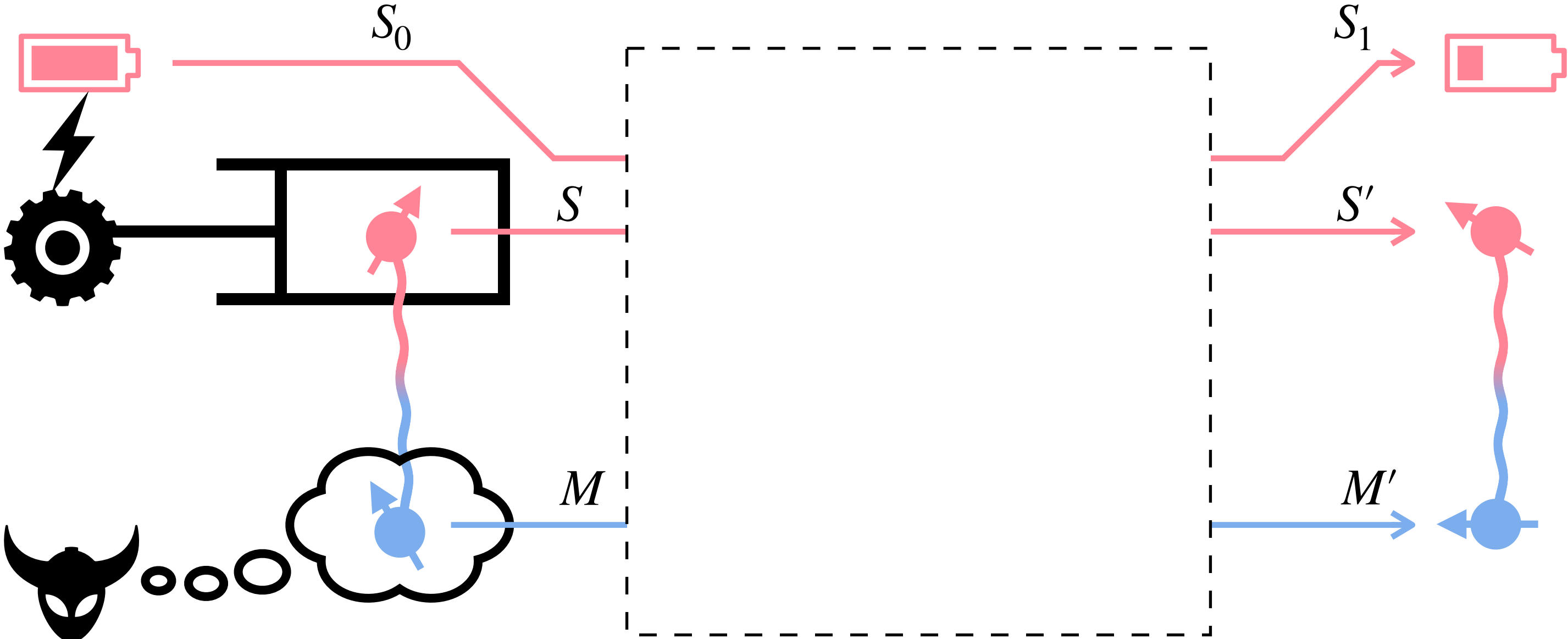
A channel $\mathcal{N}_{SM \rightarrow S'M'}$ is a QFGO if and only if

$$\mathcal{N}_{SM \rightarrow S'M'} \circ \mathcal{R}_{S \rightarrow S}^\gamma = \mathcal{R}_{S' \rightarrow S'}^{\gamma'} \circ \mathcal{N}_{SM \rightarrow S'M'}$$

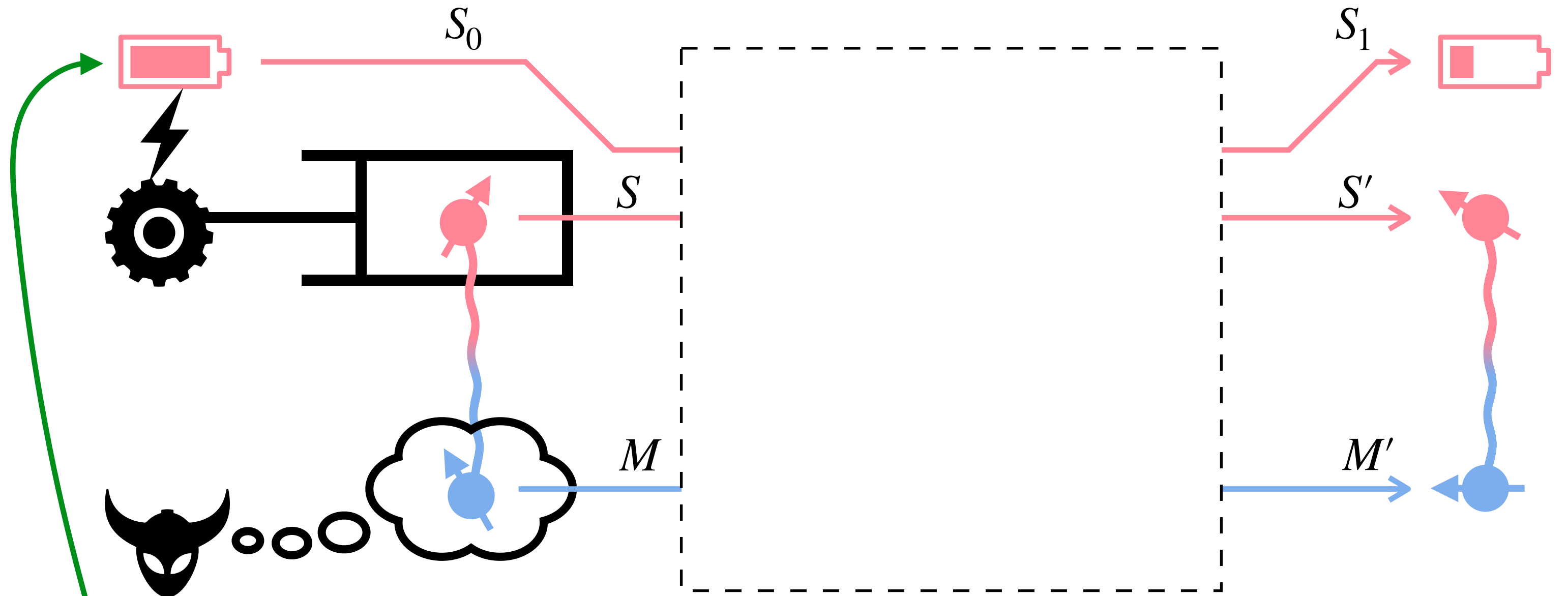
thermalization of system: $\mathcal{R}_{S \rightarrow S}^\gamma[\cdot] = \text{Tr}_S[\cdot] \gamma_S$



Work accounting



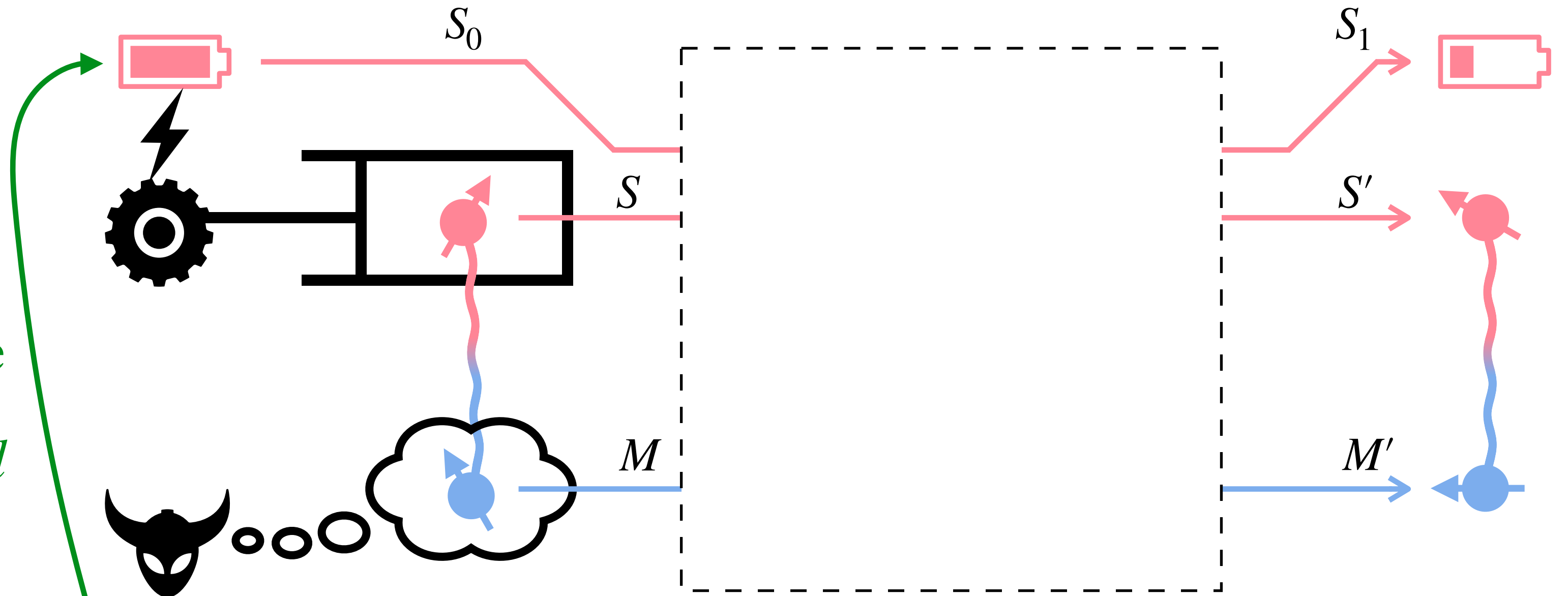
Work accounting



Allow system to exchange work with a battery

Work accounting

- Battery is always in pure state $|0\rangle\langle 0|$ and has uniform Gibbs state
- By Landauer, battery of dimension d has work storage $\beta^{-1} \ln d$



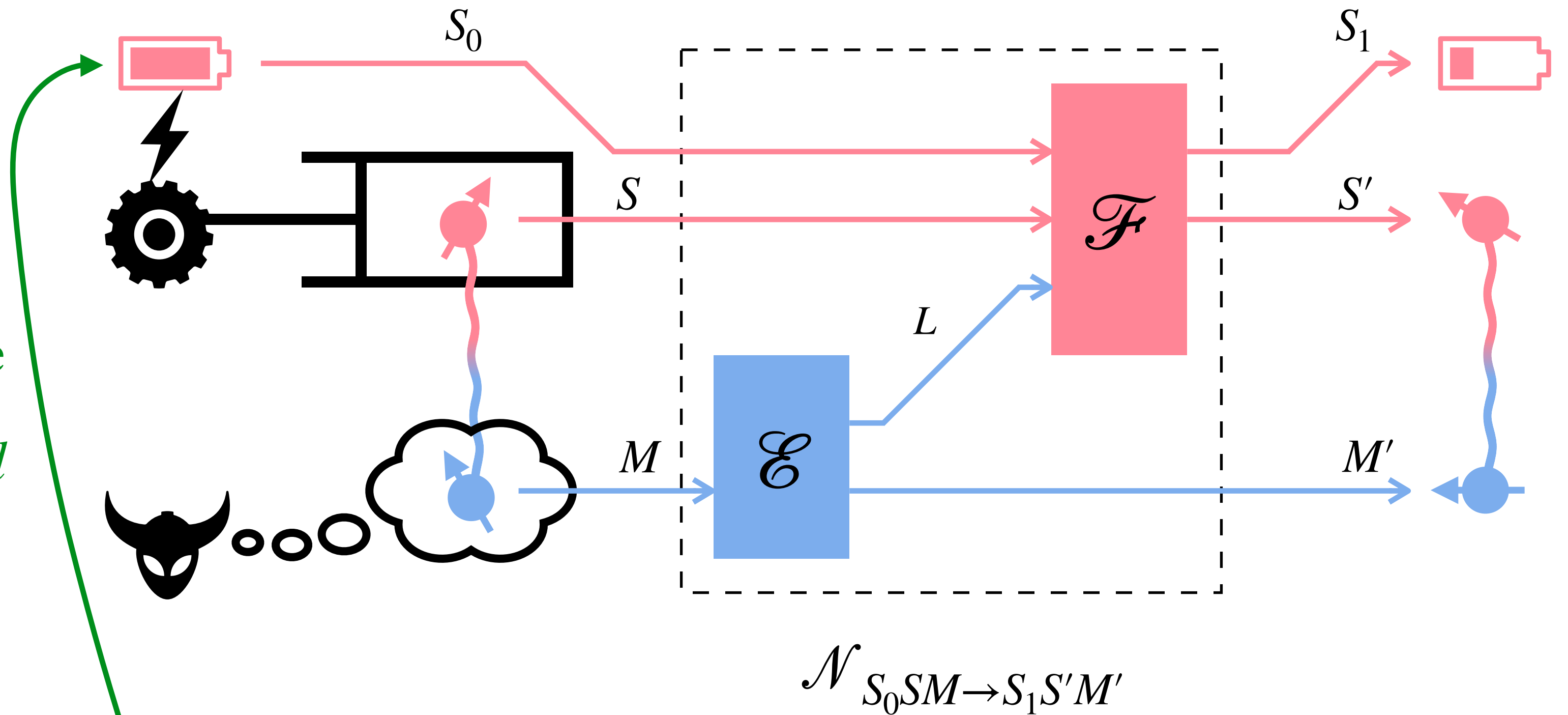
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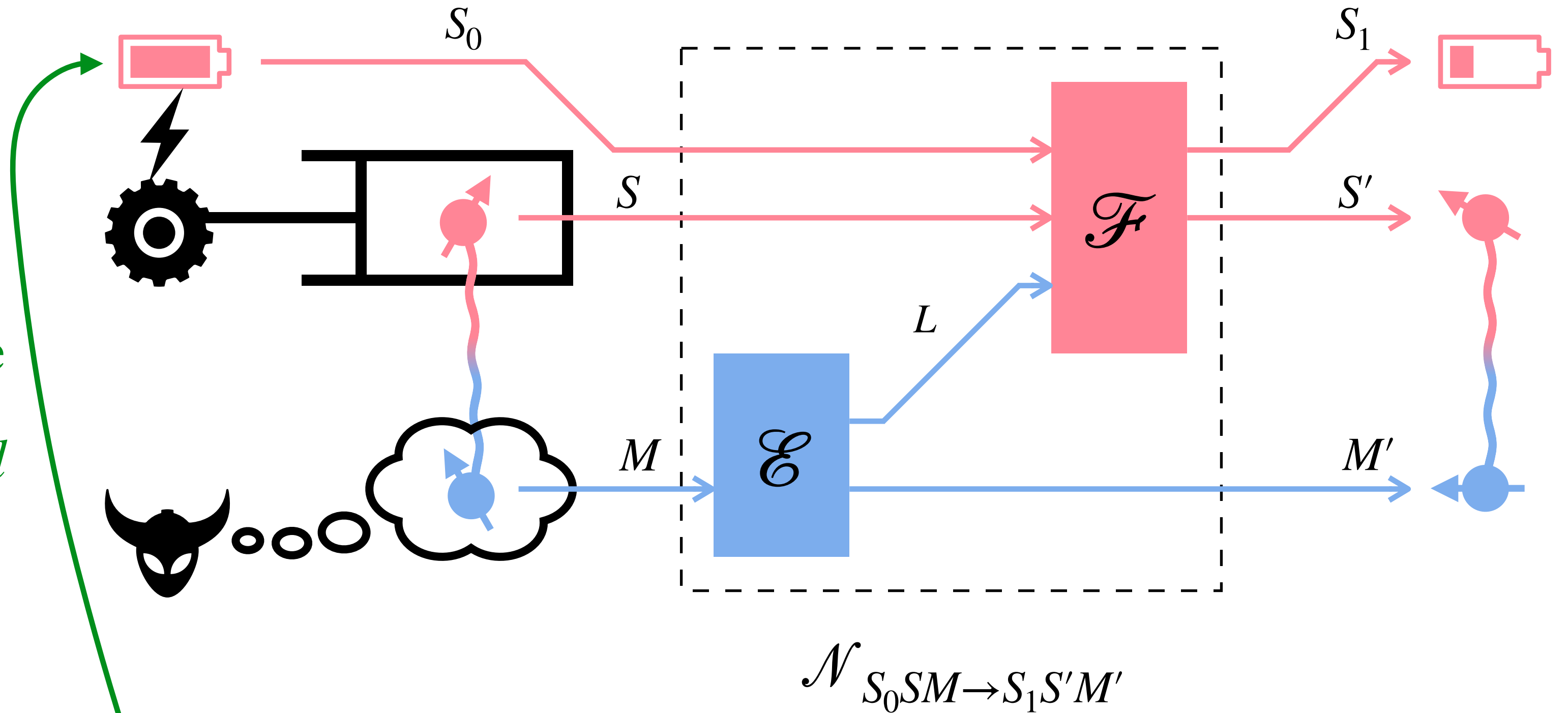
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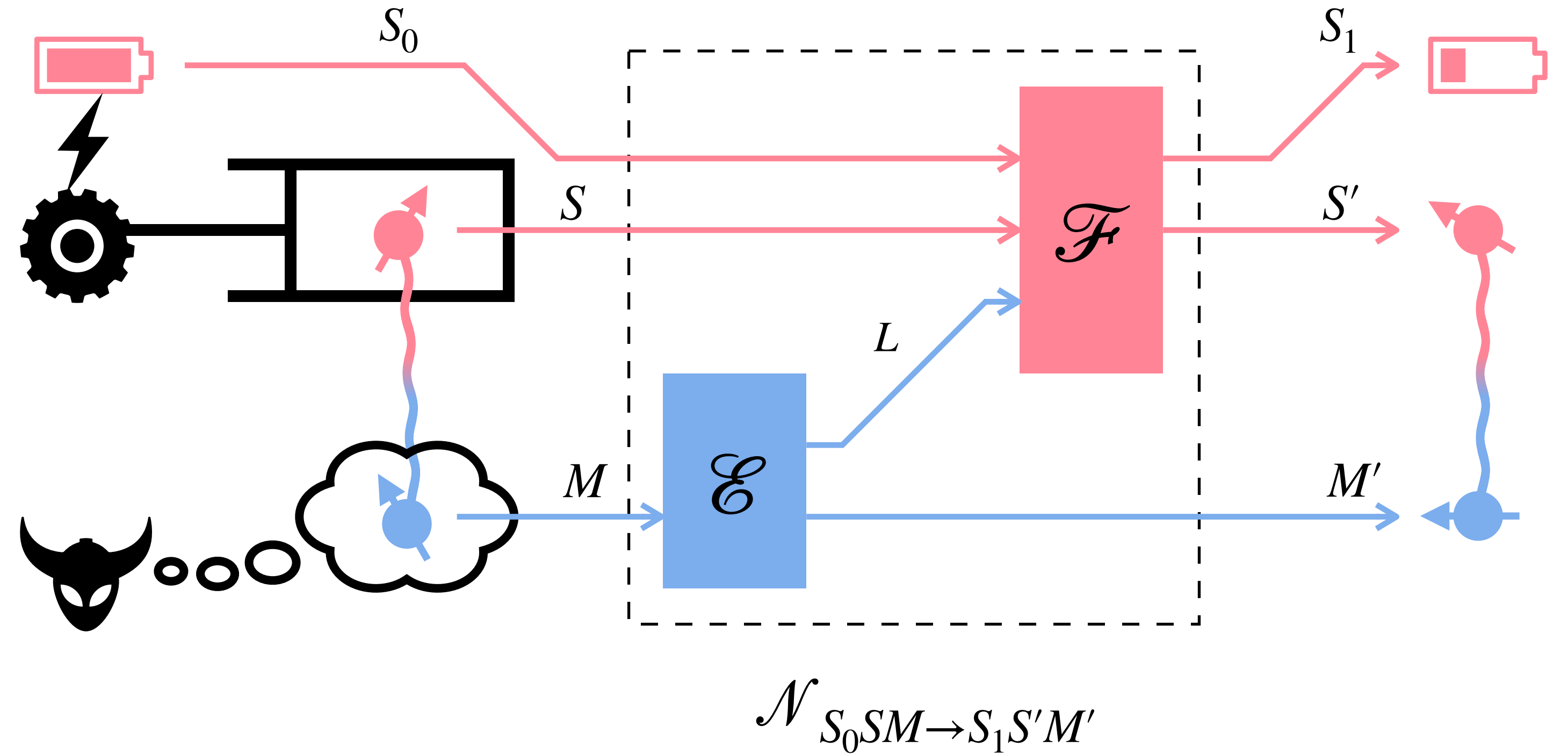
Battery-system-memory undergo free operation (QFGO)

The decrease in battery's work storage is work expended on system during conversion:

$$\beta^{-1}(\ln |S_0| - \ln |S_1|)$$

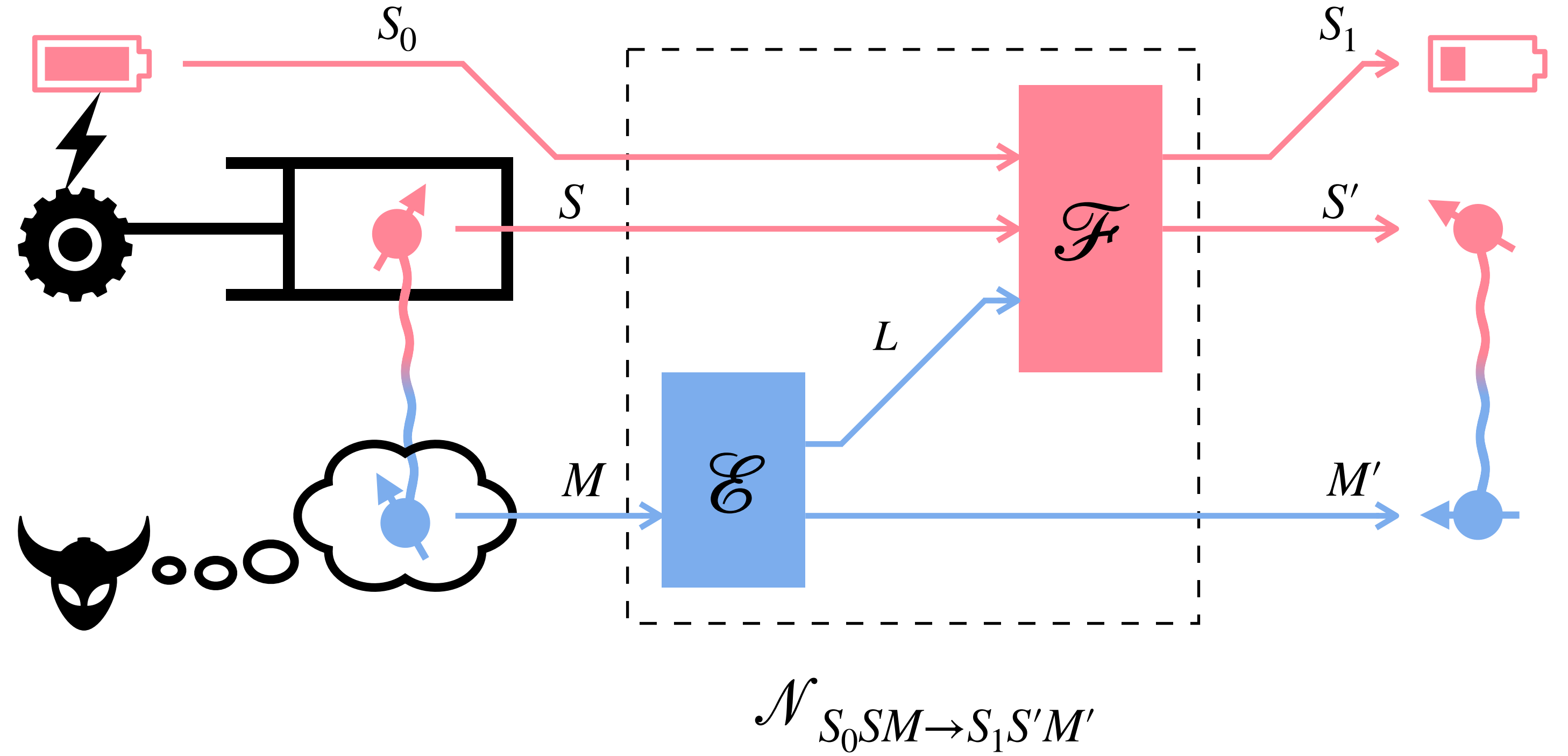
if $|S_1| < |S_0|$, work is consumed; if $|S_1| > |S_0|$, work is gained

Definition of work cost



Single-shot work cost under QFGOs: least amount of work required to convert (ρ_{SM}, γ_S) to $(\rho'_{S'M'}, \gamma'_{S'})$ up to error ε using QFGO

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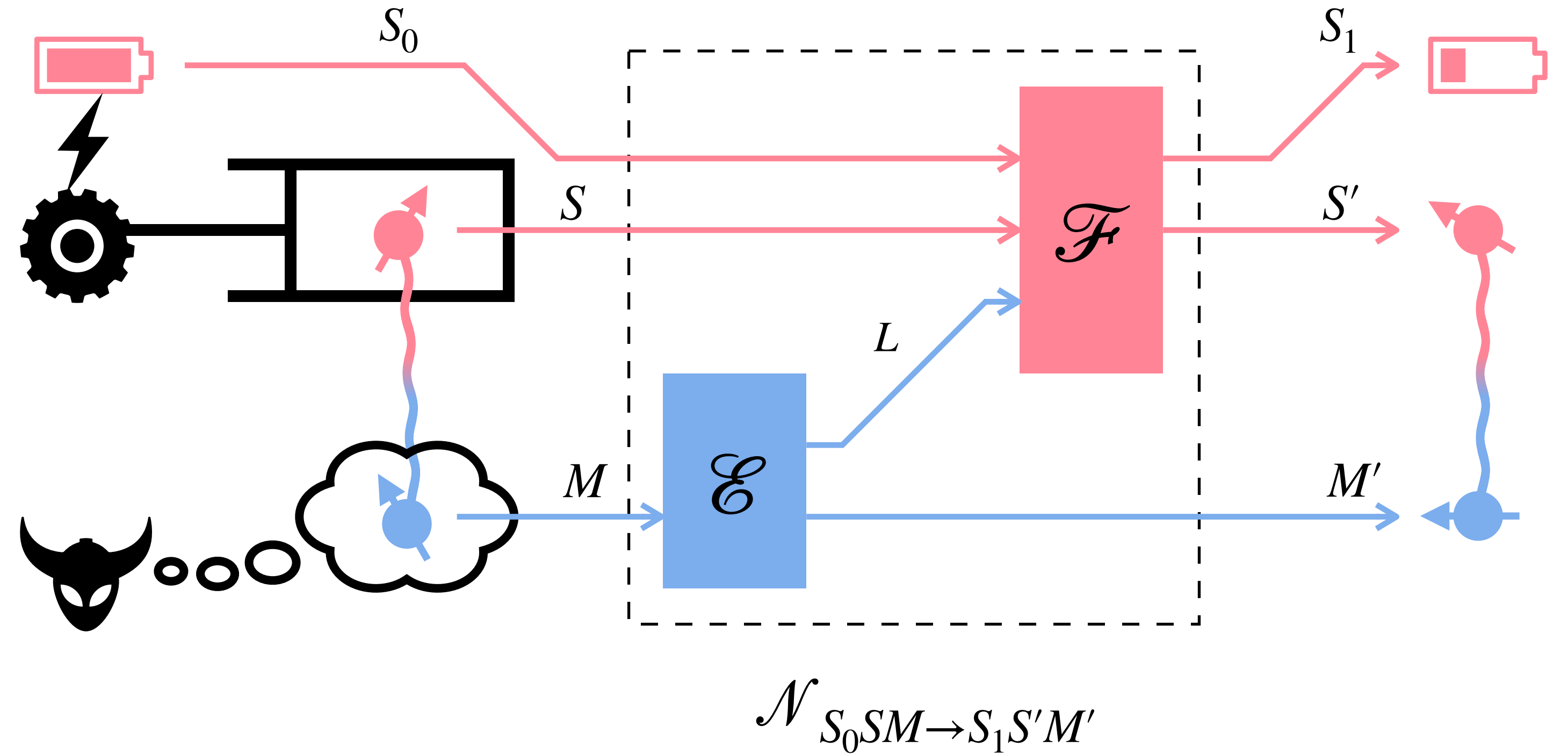


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$$W^\varepsilon((\rho_{SM}, \gamma_S) \rightarrow (\rho'_{S'M'}, \gamma'_{S'}))$$

$$= \inf_{\substack{S_0, S_1, \\ \mathcal{N} \in \text{QFGO}}} \left\{ \beta^{-1}(\ln |S_0| - \ln |S_1|) : \mathcal{N}_{S_0 S M \rightarrow S_1 S' M'}[|0\rangle\langle 0|_{S_0} \otimes \rho_{SM}] \overset{\text{in trace distance}}{\approx}_\varepsilon |0\rangle\langle 0|_{S_1} \otimes \rho'_{S'M'} \right\}$$

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Single-shot work cost under any feedback-control scheme cannot be smaller than this

Main results:

Single-shot & asymptotic work with q. feedback

Task 1: system formation with q. feedback

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Starting from thermal equilibrium, create a target state (ρ_{SM}, γ_S)

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Narasimhachar et al., PRL 122, 060101 (2019)

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Definition: single-shot work of formation under QFGOs

$$W_{\text{form}}^{\varepsilon}(\rho_{SM}, \gamma_S) := W^{\varepsilon}((\gamma_S, \gamma_S) \rightarrow (\rho_{SM}, \gamma_S))$$

Task 1: system formation with q. feedback

Theorem. Single-shot & asymptotic work of formation under QFGOs:

$$W_{\text{form}}^{\varepsilon}(\rho_{SM}, \gamma_S) = \beta^{-1} I_{\text{max}}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in [0, 1]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} W_{\text{form}}^{\varepsilon}(\rho_{SM}^{\otimes n}, \gamma_S^{\otimes n}) = \beta^{-1} I(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in (0, 1)$$

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Smoothed max-generalized mutual information:

$$I_{\text{max}}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) := \inf_{t, \tau_{SM}, \sigma_M} \{ \ln t : \tau_{SM} \leq t \gamma_S \otimes \sigma_M, \tau_M \leq \sigma_M, \tau_{SM} \approx_{\varepsilon} \rho_{SM} \}$$

τ_{SM} is subnormalized state, σ_M is normalized state

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Umegaki generalized mutual information: τ_{SM} is subnormalized state, σ_M is normalized state

$$I(\rho_{SM} \| \gamma_S) := D(\rho_{SM} \| \gamma_S \otimes \rho_M) = D(\rho_S \| \gamma_S) + I(S; M)_\rho$$

Task 1: system formation with q. feedback

Theorem. Single-shot & asymptotic work of formation under QFGOs:

$$W_{\text{form}}^{\varepsilon}(\rho_{SM}, \gamma_S) = \beta^{-1} I_{\text{max}}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in [0, 1]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} W_{\text{form}}^{\varepsilon}(\rho_{SM}^{\otimes n}, \gamma_S^{\otimes n}) = \beta^{-1} I(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in (0, 1)$$

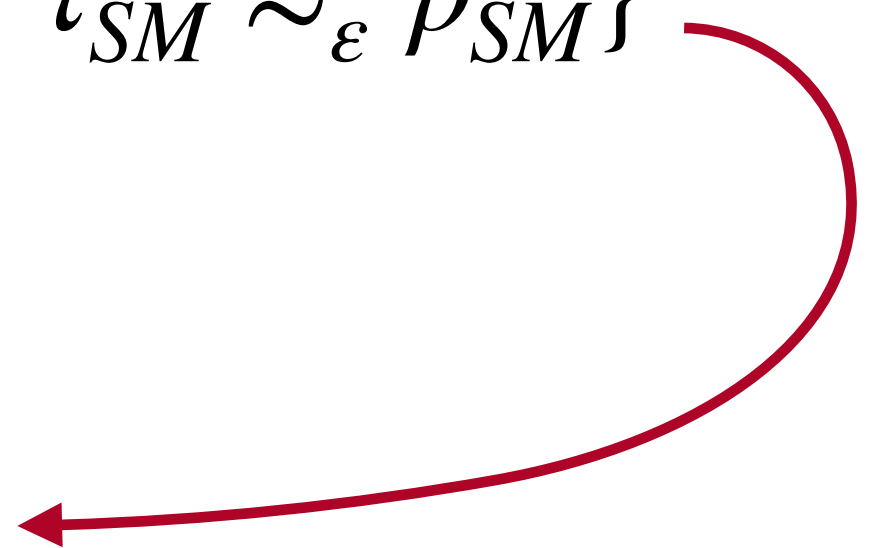
Smoothed max-generalized mutual information: **SDP computable**

$$I_{\text{max}}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) := \inf_{t, \tau_{SM}, \sigma_M} \{ \ln t : \tau_{SM} \leq t \gamma_S \otimes \sigma_M, \tau_M \leq \sigma_M, \tau_{SM} \approx_{\varepsilon} \rho_{SM} \}$$

Umegaki generalized mutual information:

$$I(\rho_{SM} \| \gamma_S) := D(\rho_{SM} \| \gamma_S \otimes \rho_M) = D(\rho_S \| \gamma_S) + I(S; M)_{\rho}$$

a new asymptotic
equipartition property



More on smoothed-max GMI

$$\delta(\tau, \rho) := \frac{1}{2}(\|\tau - \rho\|_1 + |\text{Tr}[\tau] - \text{Tr}[\rho]|)$$

in generalized trace distance

$$I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) := \inf_{t, \tau_{SM}, \sigma_M} \{ \ln t : \tau_{SM} \leq t \gamma_S \otimes \sigma_M, \tau_M \leq \sigma_M, \tau_{SM} \approx_{\varepsilon} \rho_{SM} \}$$

τ_{SM} is subnormalized state, σ_M is normalized state

- If $\varepsilon = 0$: $I_{\max}^{\uparrow}(\rho_{SM} \| \gamma_S) = D_{\max}(\rho_{SM} \| \gamma_S \otimes \rho_M)$
- If $\varepsilon = 0$ and $\gamma_S = I_S$: $I_{\max}^{\uparrow}(\rho_{SM} \| I_S) = -H_{\min}^{\downarrow}(S | M)_{\rho}$

unoptimized conditional min-entropy

What's new about it?

- A different kind of smoothing: designed to ensure robustness in the presence of battery:

$$I_{\max}^{\uparrow, \varepsilon}(|0\rangle\langle 0|_{S_1} \otimes \rho_{SM} \| I_{S_1} / |S_1| \otimes \gamma_S) = I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) + \ln |S_1|$$

in standard trace distance

- Alternative expression: $I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) = \inf_{S_1, \omega_{S_1 SM}} \{ I_{\max}^{\uparrow}(\omega_{S_1 SM} \| I_{S_1} \otimes \gamma_S) : \omega_{S_1 SM} \approx_{\varepsilon} |0\rangle\langle 0|_{S_1} \otimes \rho_{SM} \}$

normalized state

Proof sketch of single-shot work of formation under QFGOs

Converse: smoothed max-GMI is

(i) resource monotone under QFGOs

(ii) robust in the presence of battery:

$$I_{\max}^{\uparrow, \epsilon}(|0\rangle\langle 0|_{S_1} \otimes \rho_{SM} \| I_{S_1} / |S_1| \otimes \gamma_S) = \ln |S_1| + I_{\max}^{\uparrow, \epsilon}(\rho_{SM} \| \gamma_S)$$

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Alternative expression of smoothed max-GMI:

$$I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) = \inf_{S_1, \omega_{S_1 SM}} \{ I_{\max}^{\uparrow}(\omega_{S_1 SM} \| I_{S_1} / |S_1| \otimes \gamma_S) - \ln |S_1| : \omega_{S_1 SM} \overset{\text{in standard trace distance}}{\approx}_{\varepsilon} |0\rangle\langle 0|_{S_1} \otimes \rho_{SM} \}$$

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Achievability: by the following QFGO

$$\mathcal{N}_{S_0 S \rightarrow S_1 SM}[\cdot] = \text{Tr}[|0\rangle\langle 0|_{S_0}(\cdot)_{S_0 S}] \omega_{S_1 SM} + \text{Tr}[(I_{S_0} - |0\rangle\langle 0|_{S_0})(\cdot)_{S_0 S}] \frac{|S_0| I_{S_1} / |S_1| \otimes \gamma_S \otimes \omega_M - \omega_{S_1 SM}}{(|S_0| - 1)}$$

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- Saturates the converse:

$$\beta^{-1}(\ln |S_0| - \ln |S_1|) = \beta^{-1} I_{\max}^{\uparrow}(\omega_{S_1 SM} \| I_{S_1} / |S_1| \otimes \gamma_S) - \ln |S_1| = \beta^{-1} I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S)$$

Task 2: work extraction with q. feedback

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Morris et al., PRL 122, 130601 (2019)

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Definition: single-shot extractable work under QFGOs

$$W_{\text{extr}}^\varepsilon(\rho_{SM}, \gamma_S) := - W^\varepsilon((\rho_{SM}, \gamma_S) \rightarrow (\gamma_S, \gamma_S))$$

Task 2: work extraction with q. feedback

Theorem. Single-shot & asymptotic extractable work under QFGOs:

$$W_{\text{extr}}^{\varepsilon}(\rho_{SM}, \gamma_S) = \beta^{-1} I_{\min}^{\downarrow, \varepsilon}(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in [0, 1]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} W_{\text{extr}}^{\varepsilon}(\rho_{SM}^{\otimes n}, \gamma_S^{\otimes n}) = \beta^{-1} I(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in (0, 1)$$

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$$\begin{aligned} I_{\min}^{\downarrow, \varepsilon}(\rho_{SM} \| \gamma_S) &:= \sup_{\Lambda_{SM}} \{ -\ln \|\text{Tr}_S[\Lambda_{SM} \gamma_S]\|_\infty : \text{Tr}[\Lambda_{SM} \rho_{SM}] \geq 1 - \varepsilon, 0 \leq \Lambda_{SM} \leq I_{SM} \} \\ &= \inf_{\sigma_M} D_{\min}^\varepsilon(\rho_{SM} \| \gamma_S \otimes \sigma_M) \end{aligned}$$

 hypothesis-testing relative entropy

Task 2: work extraction with q. feedback

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Special case: if $\gamma_S = I_S / |S|$ (trivial Hamiltonian), then single-shot work cost of erasure

$$\begin{aligned} W^{\varepsilon}((\rho_{SM}, I_S / |S|) \rightarrow (|0\rangle\langle 0|_S, I_S / |S|)) &= -W_{\text{extr}}^{\varepsilon}(\rho_{SM}, I_S / |S|) + \beta^{-1} \ln |S| \\ &= \beta^{-1} H_{\max}^{\varepsilon}(S | M)_{\rho} \end{aligned}$$

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unlike classical case, q. conditional entropies can be negative
cooling instead of heating during erasure in the presence of q. side info.

Proof sketch of single-shot extractable work under QFGOs

Converse: smoothed min-GMI is also

- (i) resource monotone under QFGOs
- (ii) robust in the presence of battery

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↓ ↘

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$\downarrow \quad \searrow$

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Λ_{SM}

optimal measurement operator

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Generalized 2nd law of thermodynamics with q. feedback

Theorem. Asymptotic work cost under QFGOs for a general conversion:

$$\lim_{n \rightarrow \infty} \frac{1}{n} W^\varepsilon((\rho_{SM}^{\otimes n}, \gamma_S^{\otimes n}) \rightarrow (\rho_{S'M'}^{\otimes n}, \gamma_{S'}^{\otimes n})) = \beta^{-1} (I(\rho_{S'M'}' \| \gamma_{S'}') - I(\rho_{SM} \| \gamma_S))$$

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Can also be interpreted in terms of a conditional Helmholtz free energy

Bera et al., Nat. Comm. 8, 2180 (2017)

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Can also be interpreted in terms of a conditional Helmholtz free energy

Bera et al., Nat. Comm. 8, 2180 (2017)

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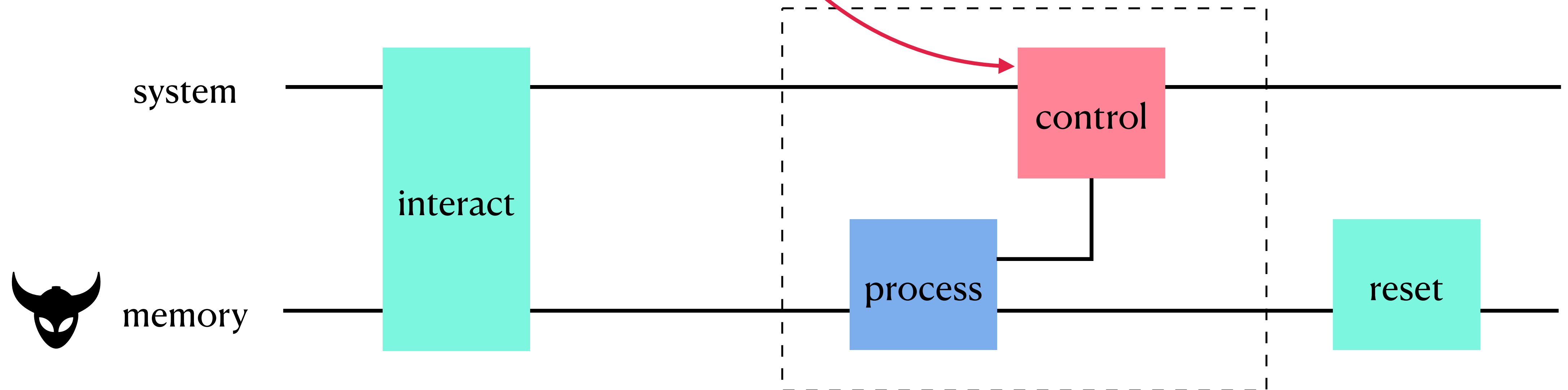
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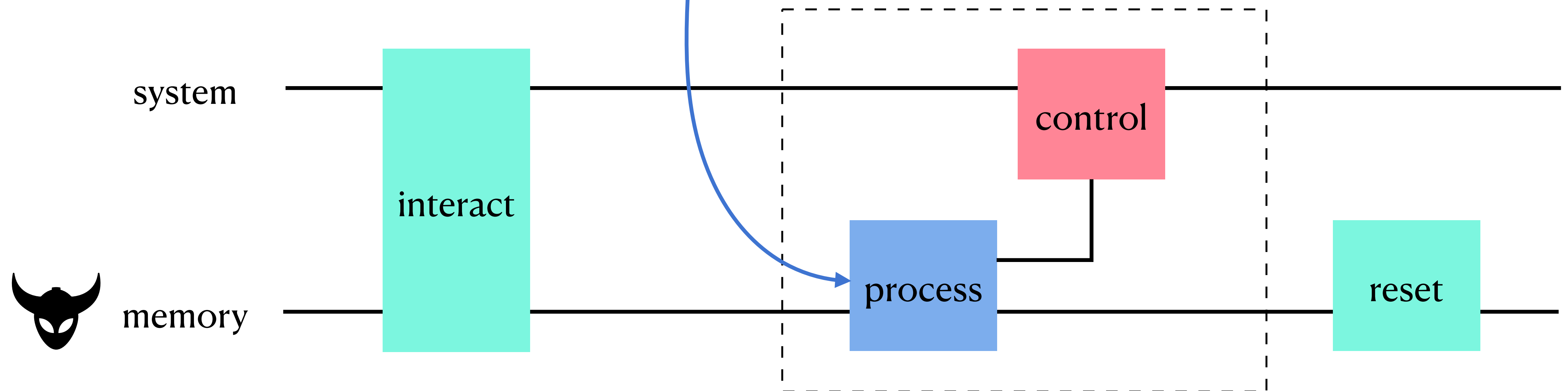
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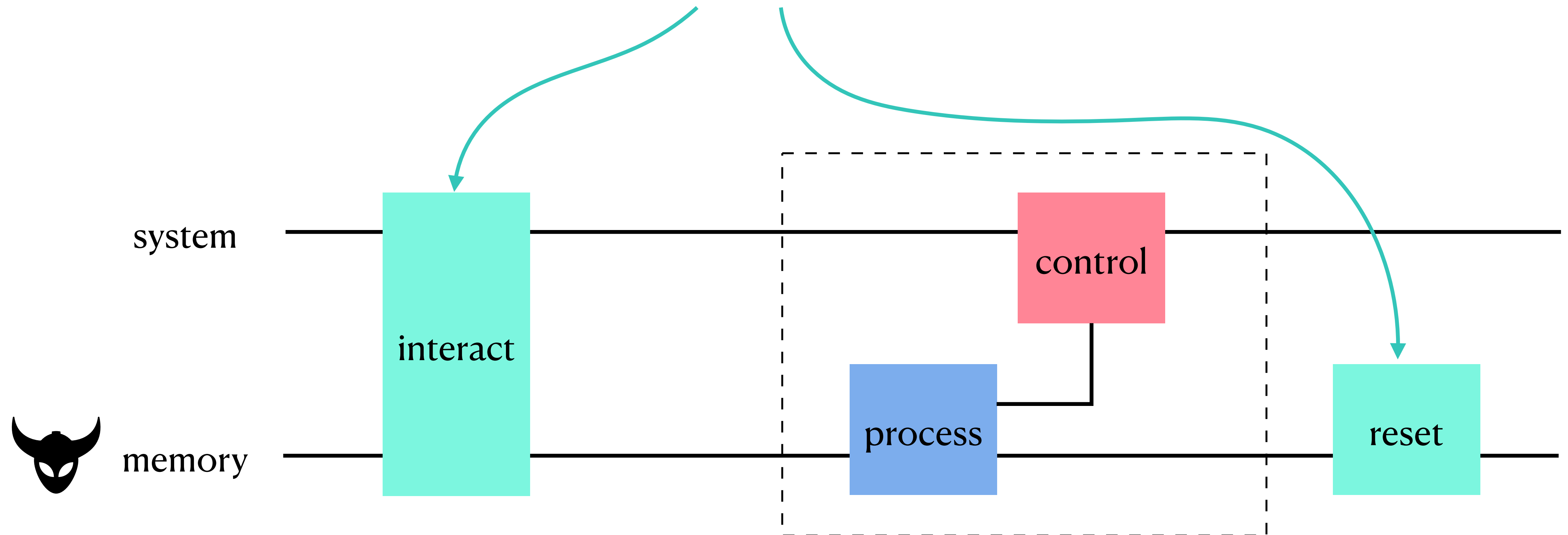
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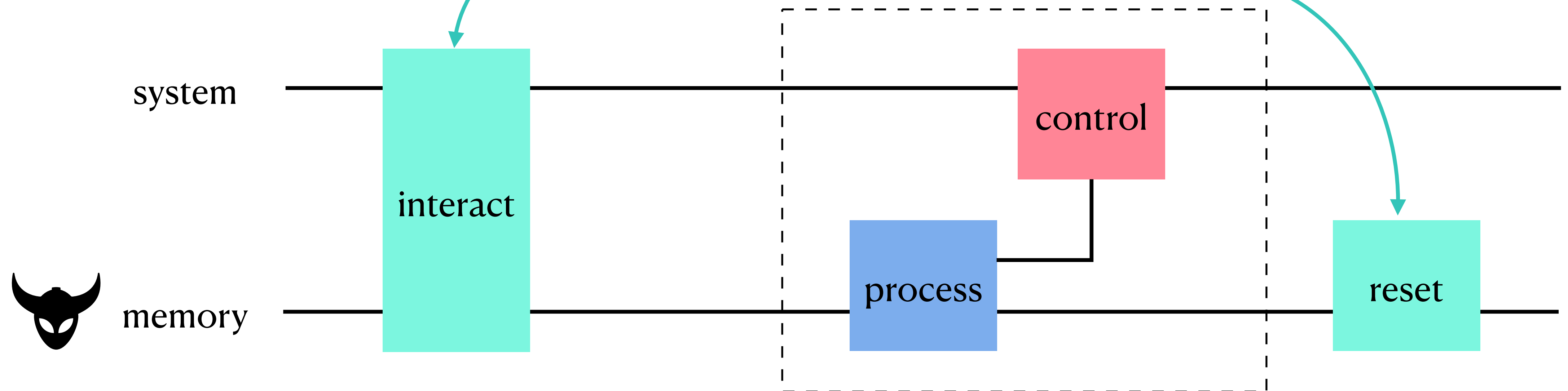
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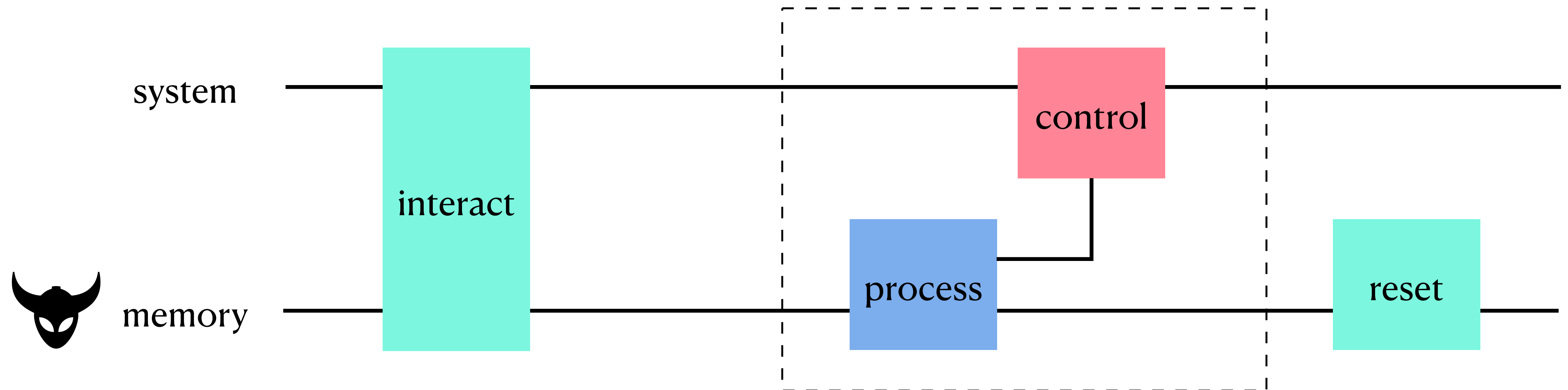
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$$\Rightarrow \langle W \rangle + \langle W_{\text{mem}} \rangle + \langle W_{\text{int\&res}} \rangle \geq 0$$



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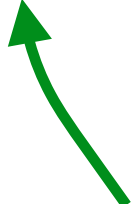
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Information theory:

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Information theory:

- Smoothed max- and min-GMI: monotone under QFGOs, robust in the presence of battery, asymptotic equipartition property
- motivates a different way of smoothing
- 

Future directions

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